

FIRST TERM – JSS 1 LIVESTOCK FARMING

WEEK 1: INTRODUCTION TO LIVESTOCK FARMING

Introduction

Livestock farming is one of the oldest and most important agricultural activities in the world. It involves the rearing of animals for human use, either for food, labour, or other valuable products. In many communities, livestock farming provides income, food, and employment, making it a vital part of both rural and urban life.

WHAT IS LIVESTOCK FARMING?



PRODUCTS INCLUDE: MILK, MEAT, EGGS, AND FERTILIZER

Meaning of Livestock Farming

Livestock farming is the practice of rearing or keeping animals such as cattle, goats, sheep, pigs, rabbits, and poultry for various purposes such as meat, milk, eggs, hides, and manure.

It is also called **animal husbandry**.

In simple terms, livestock farming means the breeding, feeding, and management of domestic animals for human benefit.

Examples of Livestock Animals

Livestock animals can be grouped according to their type and use.

Common examples include:

1. **Cattle** – for meat, milk, hides, and farm work
2. **Goats** – for meat, milk, and leather
3. **Sheep** – for meat, wool, and milk
4. **Pigs** – for pork and lard (fat)
5. **Rabbits** – for meat and fur
6. **Poultry (birds)** – for eggs and meat
7. **Fish** – for food and commercial purposes
8. **Donkeys and Horses** – for transport and labour

Importance of Livestock Farming

Livestock farming plays a key role in the economy, health, and food security of any nation. The major importance includes:

1. **Food Production:** Provides meat, milk, eggs, and other nutritious foods.
2. **Income Generation:** Farmers earn money by selling animals and animal products.

3. **Employment:** Creates job opportunities for farmers, herders, veterinarians, and feed suppliers.
4. **Raw Materials for Industry:** Provides leather, wool, bones, and fat used in industries.
5. **Source of Manure:** Animal waste (dung) is used to fertilize farmlands.
6. **Cultural and Social Importance:** Animals are used for ceremonies, dowries, and traditional events.
7. **Transportation:** Animals such as horses, donkeys, and camels are used for carrying loads.
8. **Wealth and Savings:** Livestock are considered a form of savings or investment in rural communities.

Types of Livestock Animals

Livestock animals are categorized based on their kind and the purpose for which they are kept:

Category	Examples	Main Products
Ruminants (animals with four stomachs)	Cattle, goats, sheep	Milk, meat, hides, manure
Non-ruminants (animals with one stomach)	Pigs, rabbits, poultry	Meat, eggs, fat
Aquatic animals	Fish, snails	Protein food, commercial sale
Working animals	Horses, donkeys, camels	Transportation, farm labour

Benefits of Livestock to Humans

Livestock animals are beneficial to humans in several ways:

1. They provide **food** such as meat, milk, and eggs.
2. They serve as a **source of income** for farmers.
3. They offer **raw materials** for industries (hides, wool, bones, horns).
4. They provide **organic manure** for crop farming.
5. They are used for **transportation and labour** in rural areas.
6. They contribute to **cultural and traditional practices**.
7. They help improve **nutrition and health** through animal protein.
8. They provide **employment** opportunities across the production chain.

Problems Facing Livestock Farming in Nigeria

Although livestock farming is beneficial, it faces some challenges, including:

1. **Shortage of feed and water.**
2. **Diseases and parasites** affecting animals.
3. **Poor housing and management practices.**
4. **Inadequate veterinary services.**
5. **Poor marketing and transportation facilities.**
6. **Lack of capital and modern technology.**
7. **Conflict between farmers and herders.**

Ways to Improve Livestock Farming

To make livestock farming more productive, farmers should:

1. Provide adequate feed and clean water.
2. Build proper housing for animals.

3. Vaccinate and deworm animals regularly.
4. Keep records of animals and their health.
5. Use improved breeds for higher production.
6. Practice hygiene and good sanitation.
7. Seek veterinary and extension services.

Summary

Livestock farming is the rearing of domestic animals such as cattle, goats, sheep, pigs, and poultry for human use.

It is important for food, income, employment, and raw materials.

Although there are challenges such as disease and poor management, proper feeding, housing, and veterinary care can improve livestock production and make it more profitable.

EVALUATION:

1. Define livestock farming.
2. Mention five examples of livestock animals.
3. List five importance of livestock farming.
4. State three problems facing livestock farmers.
5. Mention four ways of improving livestock farming.

ASSIGNMENT:

1. Explain in detail what livestock farming means.
2. Classify livestock animals into ruminants and non-ruminants, giving two examples of each.

3. List and explain any five benefits of livestock farming to humans.
4. Identify and describe three major challenges faced by livestock farmers in Nigeria.
5. Suggest four possible ways of solving these challenges.



WEEK 2: TYPES OF POULTRY BIRDS

Introduction

Poultry farming is one of the most important aspects of livestock production. It provides eggs, meat, feathers, and manure, and serves as a major source of income for many farmers.

The term *poultry* refers to all domesticated birds that are raised mainly for their meat, eggs, or feathers. These birds can be kept at home, in small farms, or on large commercial farms.

This lesson focuses on the **meaning of poultry**, **common types of poultry birds**, and **the uses of poultry**.

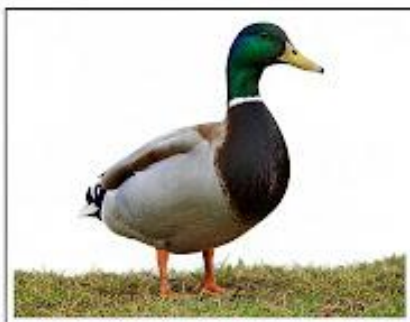
COMMON TYPES POULTRY BIRDS



CHICKEN



TURKEY



DUCK



GUINEA FOWL

Meaning of Poultry

Poultry refers to all domesticated birds that are kept by humans for food or commercial purposes such as meat, eggs, or feathers.

They are reared for both subsistence and large-scale commercial farming.

Examples include **chickens, turkeys, ducks, guinea fowls, pigeons, and quails.**

Characteristics of Poultry Birds

1. They are domesticated and cannot survive easily in the wild.
2. They are covered with feathers.
3. They lay eggs for reproduction.
4. They have two legs and two wings.
5. They feed on grains, seeds, and small insects.
6. They reproduce by laying eggs that hatch into chicks.
7. They live and move in flocks.

Common Types of Poultry Birds

1. Chicken

- The most common and widely kept poultry bird.
- Reared for **meat (broilers)** and **eggs (layers)**.
- Average lifespan is 5–10 years.
- A female chicken is called a *hen*, a male is a *cock* or *rooster*, and a young one is a *chick*.

Uses:

- Provides meat and eggs.
- Feathers can be used for decoration.

2. Turkey

- Larger than chickens.
- A male turkey is called a *tom*, a female is a *hen*, and the young one is a *poult*.
- Turkeys are reared mainly for meat and are popular during festive seasons.

Uses:

- Meat source (highly nutritious).
- Feathers used for ornamental purposes.

3. Duck

- Ducks are water-loving birds with webbed feet.
- A male duck is called a *drake*, a female is a *duck*, and the young one is a *duckling*.
- They can live both on land and in water.

Uses:

- Produce meat and eggs.
- Feathers are used for making pillows and jackets.

4. Guinea Fowl

- Native African bird known for its spotted feathers.
- A male is called a *cock*, a female is a *hen*, and the young one is a *keet*.

Uses:

- Provides tasty and lean meat.
- Serves as an alarm bird because of its loud noise.
- Acts as a pest control bird by eating insects and worms.

5. Pigeon

- Small bird that can be tamed or kept in lofts.
- A male pigeon is called a *cock*, a female a *hen*, and the young one a *squab*.

Uses:

- Provides meat.
- Used for recreation and messenger purposes in the past.

6. Quail

- A small-sized poultry bird.
- Reared mainly for meat and eggs.
- A male quail is called a *cock*, a female a *hen*, and a young one a *chick*.

Uses:

- Produces high-quality meat and eggs.
- Eggs are medicinal and nutritious.

Other Poultry Birds

- **Ostrich** – Reared for meat, leather, and feathers.
- **Peacock** – Reared mainly for beauty and ornamental purposes.
- **Geese** – Reared for meat, eggs, and feathers.

Uses and Importance of Poultry Birds

1. **Source of Food:** Provides meat and eggs rich in protein.
2. **Source of Income:** Farmers sell birds, eggs, and feathers to make money.
3. **Employment:** Poultry farming creates jobs for workers, feed sellers, and transporters.
4. **Raw Materials:** Feathers and droppings are used for making products and as manure.
5. **Cultural Purposes:** Birds are used for gifts, festivals, and sacrifices.
6. **Pest Control:** Some poultry birds, like guinea fowls, feed on insects.
7. **Manure Production:** Poultry droppings enrich the soil for crop farming.

Factors to Consider When Choosing a Type of Poultry to Rear

1. **Purpose of rearing** – meat, egg, or ornamental use.
2. **Available capital** – amount of money to start and maintain the business.
3. **Climate** – some birds perform better in certain environments.
4. **Space and housing** – available land area for rearing.
5. **Availability of feed and water.**
6. **Market demand** – type of birds that sell fast in the local market.

Summary

Poultry birds are domesticated birds reared mainly for meat, eggs, or other products.

Examples include chickens, turkeys, ducks, guinea fowls, pigeons, and quails.

They provide food, income, and raw materials for industries.

Proper selection and management of poultry types ensure successful poultry farming.

EVALUATION:

1. Define poultry.
2. List five examples of poultry birds.
3. Mention three uses of poultry birds.
4. State two characteristics of poultry birds.
5. Explain two factors to consider when choosing a type of poultry to rear.

ASSIGNMENT:

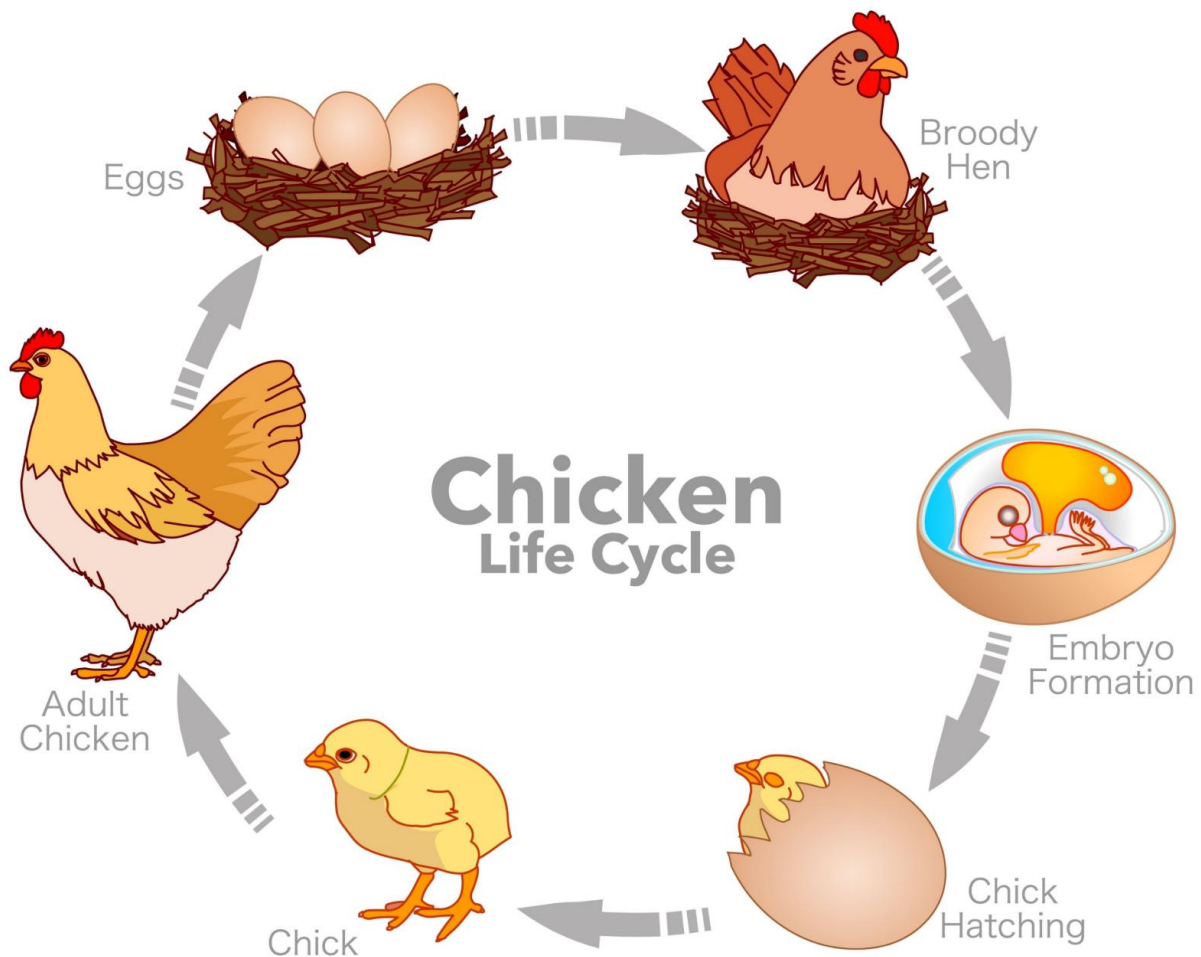
1. Describe in detail any four types of poultry birds and their uses.
2. List and explain five importance of poultry farming.
3. Draw and label any two types of poultry birds.
4. Give five differences between chicken and turkey.
5. Suggest four reasons why poultry farming is profitable in Nigeria.

WEEK 3: BREEDS OF POULTRY BIRDS

Introduction

Poultry farming involves the rearing of domesticated birds for meat, eggs, or other purposes. However, not all poultry birds are the same. Within each type—especially chickens—there are **different breeds** that possess unique physical features, productivity levels, and adaptability. Understanding poultry breeds helps farmers select the right type for egg or meat production and for suitable environmental conditions.

This lesson focuses on the **meaning of poultry breeds, types of breeds, examples of local and exotic breeds, and their characteristics.**



Meaning of Breed

A **breed** is a group of animals of the same species that have similar characteristics such as size, colour, shape, and productivity, and that pass these traits on to their offspring.

For example:

- In chickens, there are **layers** (for eggs) and **broilers** (for meat).
- Each breed has different growth rates, body size, and production capacity.

Classification of Poultry Breeds

Poultry breeds can be grouped according to different criteria:

1. According to Origin:

- **Local (Indigenous) Breeds**
- **Exotic (Foreign or Improved) Breeds**

2. According to Purpose:

- **Egg-producing breeds (Layers)**
- **Meat-producing breeds (Broilers)**
- **Dual-purpose breeds (Egg and meat producers)**

POULTRY BREEDS

LAYER



EGG PRODUCTION

BROILER



MEAT PRODUCTION

LOCAL BREEDS OF POULTRY

Meaning:

Local breeds are the traditional or native birds found in rural areas. They are not improved through modern breeding techniques. They are usually hardy and can survive under harsh conditions.

Characteristics of Local Breeds:

1. Small body size and slow growth rate.
2. Low egg and meat production.
3. Resistant to common diseases.

4. Can survive on natural feed like grains and insects.
5. Commonly reared under the *free-range system*.
6. Require little capital to manage.

Examples of Local Breeds:

- **Local fowl (village chicken)** – Common in most rural homes.
- **Local duck** – Raised near rivers and ponds.
- **Guinea fowl** – Native to Africa; known for noisy calls and dotted feathers.
- **Local turkey** – Tough and smaller than foreign ones.

EXOTIC BREEDS OF POULTRY

Meaning:

Exotic breeds are improved or imported birds from other countries. They are usually crossbred for higher productivity in eggs or meat. They require good housing, balanced feeding, and proper management.

Characteristics of Exotic Breeds:

1. Grow faster and larger than local breeds.
2. High egg and meat production.
3. Require more care, housing, and feeding.
4. Susceptible to diseases if not well managed.
5. Commonly used in commercial poultry farms.
6. Bred through scientific improvement.

COMMON EXOTIC BREEDS OF CHICKENS

1. Layers (Egg-producing breeds)

Layers are reared mainly for the production of eggs.

They start laying eggs at about 18–22 weeks of age and can lay up to 250–300 eggs per year under proper care.

Examples:

- **Isa Brown**
- **Bovan Nera**
- **Hy-Line Brown**
- **Black Australorp**
- **Leghorn**

Characteristics of Layers:

1. Light body size.
2. Active and alert in movement.
3. Produce large quantities of eggs.
4. Consume less feed compared to broilers.
5. Feathers are usually light in colour (white, brown, or black).

2. Broilers (Meat-producing breeds)

Broilers are kept mainly for meat production. They grow very fast and are ready for market within 6–8 weeks.

Examples:

- **Arbor Acres**
- **Ross 308**

- **Cobb 500**
- **Marshall**
- **Hubbard**

Characteristics of Broilers:

1. Large body size with thick muscles.
2. Fast growth rate.
3. Consume more feed for body building.
4. Produce tender, juicy meat.
5. Short production cycle (6–8 weeks).

3. Dual-Purpose Breeds (For Both Meat and Eggs)

Dual-purpose breeds can be used for both egg and meat production. They are preferred by small-scale farmers who want both benefits.

Examples:

- **Rhode Island Red**
- **Plymouth Rock**
- **Sussex**
- **New Hampshire Red**

Characteristics of Dual-Purpose Breeds:

1. Moderate egg production (150–200 eggs per year).
2. Medium body size suitable for meat.
3. Hardy and adaptable to local conditions.
4. Attractive feather colours.

5. Require moderate management and feeding.

DIFFERENCES BETWEEN LOCAL AND EXOTIC BREEDS

Feature	Local Breeds	Exotic Breeds
Growth Rate	Slow	Fast
Egg Production	Low	High
Disease Resistance	High	Low
Feeding	Can scavenge for food	Require balanced feed
Cost of Maintenance	Low	High
Management	Simple	Intensive
Environment	Adapted to local weather	May need controlled housing

IMPORTANCE OF KNOWING POULTRY BREEDS

1. Helps farmers choose the right birds for egg or meat production.
2. Ensures better farm management and feeding practices.
3. Helps in disease control and breeding programs.
4. Increases productivity and profitability.
5. Encourages crossbreeding for improved local breeds.

Summary

- A **breed** is a group of animals of the same kind with similar characteristics.
- Poultry breeds can be **local or exotic**, and **egg, meat, or dual-purpose** types.

- Local breeds are hardy but low-yielding, while exotic breeds are high-yielding but require better care.
- Farmers must choose breeds based on purpose, climate, and available resources.

EVALUATION:

1. Define the term “breed.”
2. Differentiate between local and exotic breeds of poultry.
3. List three examples each of local and exotic breeds.
4. Mention three characteristics of layers.
5. State two advantages of dual-purpose breeds.

ASSIGNMENT:

1. Explain the meaning of poultry breed and state two types of classification of breeds.
2. List five examples of exotic breeds of poultry and state their uses.
3. Make a table showing four differences between local and exotic breeds.
4. Give four characteristics of broilers.
5. Describe how a farmer can improve the productivity of local breeds.

WEEK 4: POULTRY HOUSING

Introduction

Poultry housing refers to the shelter or building where poultry birds such as chickens, ducks, turkeys, or guinea fowls are kept for rearing. A poultry house provides protection, comfort, and a controlled environment for birds to grow, produce eggs, or gain weight efficiently.

Proper housing is one of the most important factors that determine the success of a poultry enterprise. When birds are well housed, they grow faster, stay healthier, and produce better.

Meaning of Poultry Housing

Poultry housing is the **act of providing a safe, comfortable, and well-ventilated shelter** for poultry birds where they can feed, rest, and lay eggs. The house should be designed to protect birds from bad weather, diseases, pests, and predators.

POULTRY REARING SYSTEMS



DEEP LITTER SYSTEM



BATTERY CAGE SYSTEM

IMPORTANCE OF POULTRY HOUSING

1. **Protection from Weather:** A poultry house protects birds from rain, wind, sun, and cold.
2. **Safety from Predators:** It keeps away animals like snakes, rats, dogs, and hawks.
3. **Disease Control:** Proper housing reduces contact with disease-causing organisms.
4. **Better Production:** Well-housed birds grow faster and lay more eggs.
5. **Ease of Management:** Makes feeding, cleaning, and egg collection easier.
6. **Proper Ventilation:** Ensures enough air and light for the birds' health.
7. **Separation of Ages:** Helps manage chicks, growers, and layers in different houses.

TYPES OF POULTRY HOUSES

There are several types of poultry houses depending on the **age, purpose, and rearing system** of the birds.

1. Brooder House

- Used for **young chicks** (from 0 to 8 weeks old).
- It provides warmth and protection.
- Equipped with **brooders** (heat sources), feeders, and drinkers.
- The floor is covered with litter material like wood shavings.

2. Grower House

- Used for **young growing birds** (from 8 to 20 weeks).
- Larger than the brooder house.
- Birds are given more space for movement and feeding.

3. Layer House

- Used for **laying hens** that produce eggs.

- Contains **laying nests**, perches, and feeding troughs.
- Can be on the floor or raised cage system (battery cage).

4. Broiler House

- Used for **broilers**, which are raised mainly for meat.
- Spacious to allow rapid growth.
- Well ventilated to prevent heat stress.

5. Breeder House

- Used for **breeding stock** (cocks and hens kept for reproduction).
- Equipped with nests and separate compartments.

POULTRY REARING SYSTEMS AND THEIR HOUSING TYPES

1. Free Range System

- Birds move freely outdoors during the day.
- They find food naturally and return to simple shelters at night.
- Low cost, but birds are exposed to predators and diseases.

2. Deep Litter System

- Birds are kept on the floor covered with litter (wood shavings or sawdust).
- Allows easy movement and natural behaviour.
- Suitable for both broilers and layers.
- Easy to manage, but requires regular cleaning.

3. Battery Cage System

- Birds are kept in wire cages arranged in tiers.
- Commonly used for commercial egg production.

- Clean, space-saving, and makes egg collection easy.
- However, it requires high capital and careful maintenance.

4. Semi-Intensive System

- Combines free-range and deep litter methods.
- Birds are housed at night and allowed to scavenge during the day.

REQUIREMENTS OF A GOOD POULTRY HOUSE

A good poultry house should have the following features:

1. **Proper Ventilation:** There must be enough windows or openings for fresh air.
2. **Adequate Lighting:** Natural sunlight or artificial light for good egg production.
3. **Dry and Warm Interior:** Floor should be dry and protected from flooding.
4. **Good Drainage System:** Prevents waterlogging and odour.
5. **Enough Space:** Avoid overcrowding to reduce disease spread.
6. **Clean and Hygienic:** Easy to clean and disinfect regularly.
7. **Strong and Safe Structure:** Should keep away predators and thieves.
8. **Orientation:** The house should face east–west to reduce heat from the sun.
9. **Availability of Feeders and Drinkers:** Ensures easy access to food and water.
10. **Accessibility:** Should be easy for the farmer to move in and out for management.

CONSTRUCTION MATERIALS FOR POULTRY HOUSES

When building a poultry house, local and affordable materials can be used, provided they meet hygiene and safety standards. Common materials include:

- **Wood:** For frames, perches, and nest boxes.

- **Wire Mesh:** For ventilation and protection from predators.
- **Blocks or Bricks:** For walls and foundation.
- **Cement and Sand:** For floors and walls.
- **Corrugated Iron Sheets or Thatch:** For roofing.
- **Nails, Hinges, and Doors:** For fittings and fixtures.
- **Plastic or Rubber Materials:** For feeders and drinkers.

GUIDELINES FOR CONSTRUCTING A POULTRY HOUSE

1. **Choose a well-drained and dry site.**
2. **Ensure the building is raised** to prevent flooding.
3. **Provide adequate windows** for ventilation and light.
4. **Ensure the floor is concrete or hard enough** for easy cleaning.
5. **Provide enough space** — about 1 square foot per bird for chicks and more for adults.
6. **Build a fence** around the poultry house to prevent intrusion by predators.
7. **Install feeding and watering facilities** at appropriate points.
8. **Regularly clean and disinfect the house.**

SUMMARY

- Poultry housing provides shelter and safety for birds.
- Types include brooder, grower, layer, broiler, and breeder houses.
- Rearing systems include free-range, deep litter, battery cage, and semi-intensive systems.
- A good poultry house must be well ventilated, clean, and spacious.

- Construction materials can be locally sourced such as wood, wire mesh, blocks, and iron sheets.

EVALUATION:

1. Define poultry housing.
2. Mention four types of poultry houses and their uses.
3. State five requirements of a good poultry house.
4. List three systems of poultry rearing and describe one.
5. Identify four materials used in constructing a poultry house.

ASSIGNMENT:

1. Explain the importance of poultry housing in livestock farming.
2. Differentiate between the deep litter and battery cage systems.
3. State five steps to consider when constructing a poultry house.
4. Describe how ventilation affects the productivity of poultry birds.
5. List five construction materials and their functions in poultry housing.

WEEK 6: POULTRY EQUIPMENT

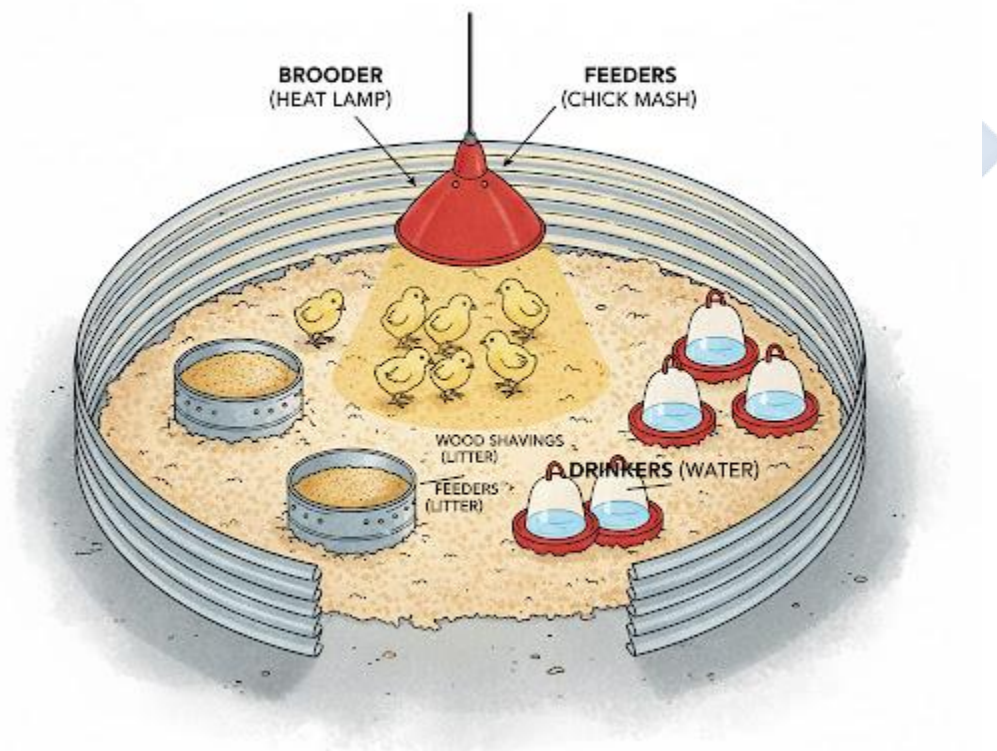
Introduction

Poultry farming requires more than just birds and feed — it also needs proper **tools and equipment** to ensure smooth management, feeding, watering, and housing of birds.

Poultry equipment refers to the **various tools, machines, and facilities used in raising and managing poultry birds efficiently.**

These equipment make the work easier, save time, and help maintain hygiene and good production. The type of equipment used depends on the scale of production — whether small (backyard), medium, or large (commercial).

BROODER HOUSE SETUP



IMPORTANCE OF POULTRY EQUIPMENT

1. **Efficient Management:** Helps the farmer to care for the birds easily and effectively.

2. **Better Feeding and Watering:** Ensures birds get enough clean feed and water.
3. **Reduced Labour:** Saves time and effort through mechanized processes.
4. **Improved Hygiene:** Keeps the poultry house neat and prevents diseases.
5. **Increased Production:** Leads to healthy and fast-growing birds.
6. **Prevents Feed Wastage:** Helps control how much feed birds consume.
7. **Protects Eggs and Birds:** Reduces damage and injury through proper handling.

TYPES OF POULTRY EQUIPMENT AND THEIR USES

Poultry equipment can be grouped into different categories based on their functions:

1. FEEDING EQUIPMENT

These are tools used to give food to poultry birds.

Common feeding equipment includes:

- **Feeders:** Containers or troughs used for supplying feed to birds. They can be made of plastic, metal, or wood.
 - *Function:* To hold and dispense feed without wastage.
 - *Types:*
 - *Manual feeders* – filled by hand.
 - *Automatic feeders* – feed birds automatically in large farms.
- **Feed Storage Bins:** Used to store feed in bulk and keep it dry and safe from pests.
 - *Function:* To preserve feed quality and prevent spoilage.

Care of Feeding Equipment:

- Clean daily to remove leftover feed.

- Protect from moisture and rodents.
- Store in a dry place.

2. WATERING EQUIPMENT

These are tools used to supply clean water to birds.

- **Drinkers:** Containers used by poultry to drink water. They come in different types and sizes depending on the age of the birds.
 - *Types:*
 - *Manual drinkers* (plastic or metal).
 - *Automatic drinkers* (nipple or bell type).
 - *Function:* To ensure birds have a constant supply of clean water.
- **Water Troughs:** Long containers used especially for larger birds.
- **Water Tanks:** Used to store water for daily use.

Care of Watering Equipment:

- Wash daily with clean water and disinfect regularly.
- Avoid algae growth and contamination.
- Ensure constant water flow in automatic systems.

3. HEATING EQUIPMENT

Used to provide warmth to chicks that cannot regulate their body temperature.

This is very important in the **brooding stage** (first 4–8 weeks).

Common heating equipment includes:

- **Brooders:** Provide controlled heat and light for chicks.

- *Types:*
 - *Coal or charcoal stove brooders.*
 - *Kerosene or gas brooders.*
 - *Electric brooders.*
- *Function:* To keep chicks warm, especially at night or during cold weather.
- **Infrared Lamps or Bulbs:** Produce both light and heat for young chicks.
- **Charcoal Pots:** Simple local heaters used in small-scale poultry farms.

Care:

- Regulate temperature to avoid overheating.
- Keep fire sources safe and away from litter materials.

4. LIGHTING EQUIPMENT

Light helps birds to feed, drink, and lay eggs properly.

Examples include:

- **Electric Bulbs or Tubes** – used for both light and heat.
- **Solar Lamps** – used where electricity is unavailable.
- **LED Lights** – energy-efficient lighting for large poultry farms.

Functions of Lighting Equipment:

1. Encourages feeding and growth, especially in chicks.
2. Improves egg production in layers.
3. Provides warmth when needed.

5. EGG HANDLING EQUIPMENT

Used in farms that produce eggs.

Examples:

- **Laying Nests:** Small boxes where hens lay eggs. They provide comfort and reduce egg breakage.
- **Egg Crates or Trays:** Used to collect and store eggs safely.
- **Egg Candles:** Used to check the quality of eggs by shining light through them.
- **Egg Incubators:** Machines that hatch eggs artificially by maintaining temperature and humidity.

Care:

- Clean nests and trays regularly.
- Disinfect incubators after each use.

6. SANITATION AND MAINTENANCE EQUIPMENT

Used to maintain cleanliness and hygiene in the poultry house.

Examples include:

- **Scrapers and Brushes:** For removing droppings and dirt.
- **Disinfectant Sprayers:** Used to apply disinfectant and kill germs.
- **Buckets and Brooms:** For general cleaning.
- **Wheelbarrows and Shovels:** For carrying litter and manure.

Importance:

- Reduces disease outbreak.
- Keeps the house neat and odour-free.
- Promotes healthy growth and productivity.

7. OTHER EQUIPMENT

- **Perches:** Wooden bars fixed inside the house for birds to rest on.
- **Ventilation Fans:** Improve air circulation in large poultry houses.
- **Feed Mill Machines:** For grinding and mixing feed ingredients.
- **Scales:** For weighing birds or feed accurately.

MAINTENANCE OF POULTRY EQUIPMENT

Proper care and maintenance ensure long-lasting use and good hygiene.

Tips for Maintenance:

1. Wash feeders and drinkers daily.
2. Disinfect equipment weekly to kill germs.
3. Repair or replace damaged tools immediately.
4. Store equipment in a clean, dry area when not in use.
5. Regularly check electrical equipment for faults.
6. Keep heating equipment safe from litter materials to prevent fire.
7. Avoid rust by cleaning and drying metal equipment after washing.

SUMMARY

- Poultry equipment are essential tools for feeding, watering, heating, lighting, and maintaining poultry.
- They include feeders, drinkers, brooders, incubators, and cleaning tools.
- Proper care, cleaning, and maintenance improve productivity and reduce losses.

- The choice of equipment depends on the size of the poultry farm and the age of the birds.

EVALUATION:

1. Define poultry equipment.
2. Mention five examples of poultry equipment and their uses.
3. List three heating equipment and state their importance.
4. Explain why sanitation equipment is important in poultry farming.
5. Describe two ways of maintaining poultry equipment.

ASSIGNMENT:

1. State five functions of poultry equipment.
2. Write short notes on feeders, drinkers, and brooders.
3. Differentiate between manual and automatic feeders.
4. List four examples of egg handling equipment and their uses.
5. Suggest five safety measures to observe when using heating equipment.

WEEK 8: POULTRY MANAGEMENT PRACTICES

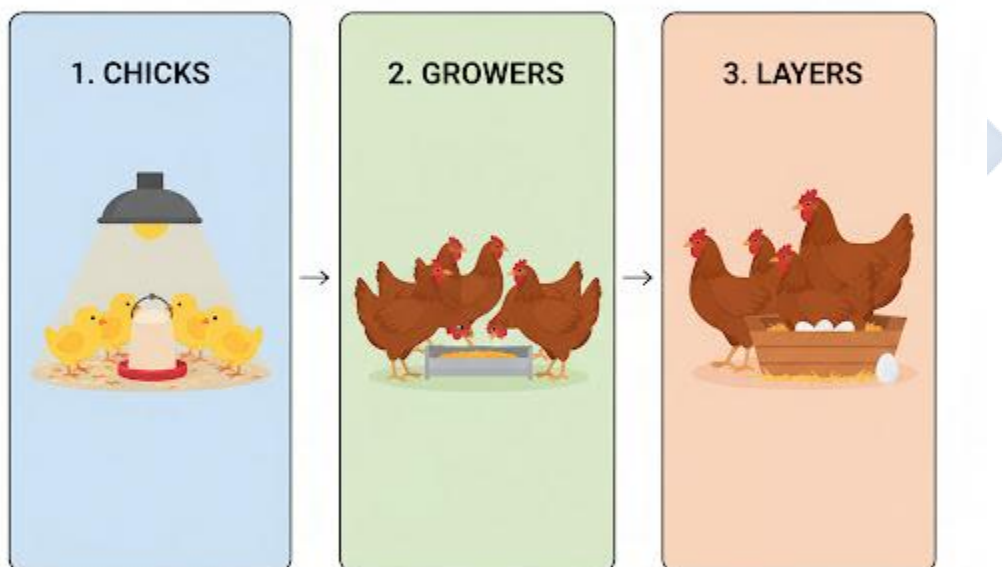
Introduction

Poultry management practices refer to all the **activities carried out to ensure that poultry birds are healthy, comfortable, and productive.**

Proper management is essential from the day-old stage (chicks) to the adult stage (layers or broilers). It includes feeding, watering, cleaning, disease prevention, and record keeping.

Good management reduces losses, increases egg and meat production, and ensures the farmer's success.

POULTRY MANAGEMENT



STAGES OF POULTRY MANAGEMENT

Poultry management practices are divided according to the **age and purpose** of the birds:

1. **Management of Chicks (0–8 weeks old)**
2. **Management of Growers (8–20 weeks old)**
3. **Management of Layers or Broilers (above 20 weeks old)**

Each stage requires special attention in feeding, housing, and care.

1. MANAGEMENT OF CHICKS (0–8 WEEKS)

Newly hatched chicks are fragile and need special care. The process of caring for them is called **brooding**.

Key Practices:

- **Brooding:** Keep chicks warm using brooders, charcoal pots, or electric bulbs. Maintain temperature between 32–35°C during the first week and reduce gradually each week.
- **Feeding:** Give chick mash (starter feed) 4–5 times daily to ensure quick growth.
- **Watering:** Provide clean, cool water at all times. Add vitamins and glucose to boost energy.
- **Litter Management:** Spread wood shavings or sawdust on the floor to keep it dry. Replace wet litter regularly.
- **Lighting:** Keep light on continuously for the first week to encourage feeding and movement.
- **Health Care:** Vaccinate chicks against common diseases (e.g., Newcastle, Gumboro).
- **Culling:** Remove weak, sick, or deformed chicks early to prevent disease spread.

Records to Keep:

- Number of chicks received
- Quantity of feed used
- Mortality record (dead birds)

- Vaccination schedule

2. MANAGEMENT OF GROWERS (8–20 WEEKS)

After brooding, chicks become **growers**. They are now stronger and require more space and less heat.

Key Practices:

- **Housing:** Transfer growers to a larger, well-ventilated house with enough feeding and drinking space.
- **Feeding:** Change their feed to grower mash to support muscle and bone development.
- **Watering:** Ensure clean water is available daily.
- **Lighting:** Provide moderate lighting for 12–14 hours daily.
- **Health Care:** Continue routine vaccination and deworming.
- **Debeaking:** Cut off the tip of the upper beak to prevent pecking and waste of feed.
- **Record Keeping:** Track weight gain, feed consumption, and health progress.
- **Segregation:** Separate cocks from hens if not needed for breeding.

Purpose of Grower Management:

- To prepare birds for egg-laying or meat production.
- To maintain healthy weight and reduce mortality.

3. MANAGEMENT OF LAYERS (EGG-PRODUCING HENS)

Layers start laying eggs from about **18–22 weeks of age**. They require special care to maintain high egg production.

Key Practices:

- **Housing:** Keep in a clean, dry, and well-lit layer house or battery cage.
- **Feeding:** Give layer mash rich in calcium and protein to aid eggshell formation and yolk quality.
- **Lighting:** Provide at least 14–16 hours of light daily to stimulate egg laying.
- **Watering:** Provide abundant clean water always — egg production depends on water availability.
- **Egg Collection:** Collect eggs 2–3 times daily to avoid breakage and contamination.
- **Health Care:** Maintain strict biosecurity; vaccinate and isolate sick birds.
- **Cleaning:** Remove droppings daily and wash feeders and drinkers regularly.
- **Culling:** Remove old or non-laying hens and replace with new stock.

4. MANAGEMENT OF BROILERS (MEAT-PRODUCING BIRDS)

Broilers are raised mainly for meat. They grow rapidly and are ready for market within 6–8 weeks.

Key Practices:

- **Housing:** Keep broilers in a spacious, well-ventilated house (deep litter system).
- **Feeding:** Provide broiler starter feed for the first 4 weeks, then finisher feed till maturity.
- **Lighting:** Provide constant light to encourage feeding.
- **Watering:** Always give clean water to aid digestion.
- **Health Care:** Vaccinate against major diseases and keep litter dry.
- **Culling:** Remove sick birds immediately to prevent spread.
- **Record Keeping:** Record feed consumption, weight gain, and mortality.

Purpose of Broiler Management:

- To produce healthy, tender, and well-fleshed meat birds quickly.

SANITATION PRACTICES IN POULTRY MANAGEMENT

Good sanitation helps prevent diseases and keeps the birds healthy.

Key Sanitation Practices:

1. Clean feeders and drinkers daily.
2. Disinfect poultry houses and equipment regularly.
3. Remove droppings and replace wet litter.
4. Keep rodents, flies, and predators away.
5. Isolate sick birds immediately.
6. Control visitors and workers entering the poultry house.

RECORD KEEPING IN POULTRY MANAGEMENT

Record keeping helps farmers monitor the performance of their flock.

Important Records Include:

- Stock record (number of birds received and sold)
- Mortality record
- Feed record
- Vaccination and medication record
- Egg production record
- Income and expenditure record

Importance of Record Keeping:

- Helps in planning and decision-making.

- Detects loss or disease problems early.
- Measures profit or loss in the business.

SUMMARY

- Poultry management practices involve feeding, watering, sanitation, health care, and record keeping.
- The stages of management include chicks, growers, layers, and broilers.
- Each stage requires specific care in housing, feeding, and lighting.
- Sanitation and record keeping are vital for disease prevention and good productivity.

EVALUATION:

1. Define poultry management practices.
2. List and explain three stages of poultry management.
3. Mention four management activities for chicks.
4. State four sanitation practices in poultry management.
5. Give three reasons why record keeping is important in poultry farming.

ASSIGNMENT:

1. Explain five management practices required for healthy chicks.
2. State five management activities for layers.
3. Mention three main feeds given to birds at different stages and their uses.
4. Describe how good sanitation improves poultry production.
5. Outline four records a poultry farmer should keep and explain their importance.

WEEK 9: POULTRY DISEASES AND CONTROL

Introduction

Poultry diseases are **abnormal conditions** that affect the health, growth, or productivity of birds such as chickens, turkeys, ducks, and guinea fowls.

Diseases in poultry can be caused by **germs, parasites, poor hygiene, bad feeding, or poor housing conditions**.

When birds are sick, they eat less, grow slowly, lay fewer eggs, or even die. Therefore, learning to recognize and control poultry diseases is essential for every poultry farmer.

POULTRY HEALTH



CAUSES OF POULTRY DISEASES

Poultry diseases can arise from the following factors:

1. Infectious Causes:

- Caused by germs such as bacteria, viruses, and fungi.
- They spread easily from one bird to another.
- Examples: Newcastle disease, Fowl cholera, and Fowl pox.

2. Parasitic Causes:

- Caused by internal parasites (worms) or external parasites (lice, mites).
- They suck the blood or nutrients of the birds, making them weak.

3. Nutritional Causes:

- Caused by lack or shortage of essential nutrients in feed.
- Examples: Rickets (lack of calcium or vitamin D), Curled toe paralysis (lack of vitamin B2).

4. Environmental Causes:

- Poor ventilation, wet litter, overcrowding, or dirty water can stress birds and lead to diseases.

SIGNS OF DISEASE IN POULTRY

A good poultry farmer must observe birds daily to detect early signs of sickness.

Common signs include:

- Loss of appetite
- Dullness or weakness
- Drooping wings
- Ruffled feathers
- Diarrhoea or watery droppings

- Sneezing or coughing
- Sudden drop in egg production
- Lameness or paralysis
- Death of birds

COMMON POULTRY DISEASES, THEIR SYMPTOMS, PREVENTION, AND CONTROL

1. NEWCASTLE DISEASE

- **Cause:** Virus
- **Mode of Spread:** Air, water, contact with infected birds, equipment, or people
- **Symptoms:**
 - Greenish diarrhoea
 - Coughing and sneezing
 - Twisting of the neck
 - Loss of balance
 - Sudden death of many birds
- **Prevention and Control:**
 - Vaccinate birds regularly.
 - Maintain strict sanitation and quarantine new birds.
 - Burn or bury dead birds deeply.

2. FOWL POX

- **Cause:** Virus

- **Mode of Spread:** Mosquito bites or contact with infected birds
- **Symptoms:**
 - Scabby sores on comb, wattles, and eyelids
 - Difficulty in breathing and eating (in wet form)
- **Prevention and Control:**
 - Vaccinate birds against fowl pox.
 - Control mosquitoes in the poultry house.
 - Isolate infected birds.

3. COCCIDIOSIS

- **Cause:** Protozoan parasite (*Eimeria* species)
- **Mode of Spread:** Contaminated feed, litter, or water
- **Symptoms:**
 - Bloody droppings
 - Weakness and loss of appetite
 - Drooping wings
 - Sudden death in severe cases
- **Prevention and Control:**
 - Keep litter dry and clean.
 - Add coccidiostats to feed or water.
 - Clean feeders and drinkers daily.

4. FOWL CHOLERA

- **Cause:** Bacterium (*Pasteurella multocida*)
- **Mode of Spread:** Contaminated feed, water, or contact with infected birds
- **Symptoms:**
 - Swollen wattles and joints
 - Greenish diarrhoea
 - Sudden death of healthy birds
- **Prevention and Control:**
 - Vaccinate against fowl cholera.
 - Maintain hygiene and disinfect regularly.
 - Remove and bury dead birds properly.

5. INFECTIOUS BRONCHITIS

- **Cause:** Virus
- **Mode of Spread:** Airborne (sneezing, coughing, and contaminated air)
- **Symptoms:**
 - Sneezing and nasal discharge
 - Difficulty in breathing
 - Drop in egg production and poor egg quality
- **Prevention and Control:**
 - Vaccinate birds at the right age.
 - Avoid overcrowding.
 - Keep poultry houses well-ventilated.

6. INTERNAL AND EXTERNAL PARASITES

- **Internal Parasites:** Roundworms, tapeworms
- **External Parasites:** Lice, mites, fleas
- **Symptoms:**
 - Loss of weight
 - Ruffled feathers
 - Pale comb and wattles
 - Poor growth or low egg production
- **Prevention and Control:**
 - Deworm birds regularly.
 - Dust birds and poultry houses with insecticidal powder.
 - Keep litter and nests clean and dry.

GENERAL METHODS OF PREVENTING POULTRY DISEASES

1. **Vaccination:**

Administer vaccines to protect birds from deadly viral diseases.
2. **Proper Sanitation:**

Clean and disinfect poultry houses regularly.
3. **Good Feeding and Watering:**

Provide balanced feed and clean water daily.
4. **Quarantine:**

Separate new or sick birds before mixing with healthy ones.
5. **Rodent and Insect Control:**

Keep away rats, flies, and mosquitoes that can carry diseases.

6. Avoid Overcrowding:

Provide enough space and ventilation.

7. Prompt Treatment:

Treat affected birds immediately or call a veterinarian.

8. Proper Disposal of Dead Birds:

Burn or bury dead birds to prevent disease spread.

SUMMARY

- Poultry diseases can be caused by viruses, bacteria, parasites, or poor management.
- Common poultry diseases include Newcastle, Fowl Pox, Coccidiosis, and Fowl Cholera.
- Prevention is better than cure — regular vaccination, sanitation, and proper feeding are key.
- Farmers must observe birds daily to detect early signs of disease.

EVALUATION:

1. Define poultry disease.
2. List four causes of poultry diseases.
3. State five signs that show a bird is sick.
4. Mention four common poultry diseases and their causes.
5. Explain three ways to prevent the spread of poultry diseases.

ASSIGNMENT:

1. Write short notes on the following poultry diseases:
(a) Newcastle Disease

(b) Fowl Pox

(c) Coccidiosis

2. List any five general methods for preventing poultry diseases.
3. Explain why vaccination is important in poultry farming.



WEEK 10: CLASSIFICATION OF FEED INGREDIENTS

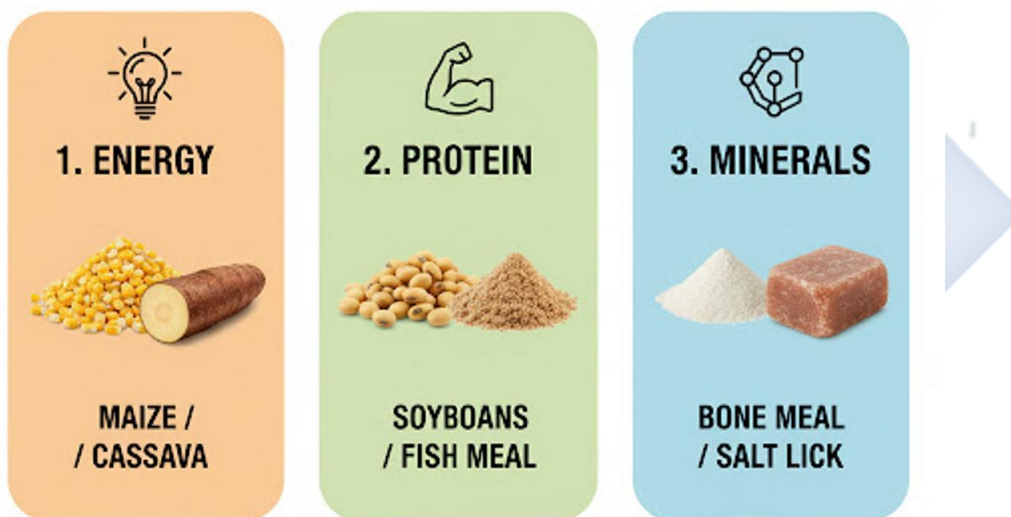
Introduction

Feeding is one of the most important aspects of livestock farming. For animals to grow well, stay healthy, and produce meat, milk, or eggs, they must be given the right kind of feed in the correct quantity.

Feed ingredients are the different materials or food items used to prepare animal feed. They supply nutrients that animals need for growth, reproduction, repair of body tissues, and energy.

A good farmer must know the different types of feed ingredients and their functions to prepare balanced feed for livestock.

WHAT'S IN AN ANIMAL FEED?



MEANING OF FEED

Feed refers to any food given to farm animals.

It can be in the form of:

- **Natural feed:** Grass, grains, crop residues.
- **Artificial feed:** Feed that is mixed or compounded by farmers or feed millers using various ingredients.

Livestock feed must contain all essential nutrients in the right proportion. These nutrients come from different **feed ingredients** that are classified into **six major groups**.

CLASSES OF FEED INGREDIENTS (NUTRIENTS)

1. Carbohydrates

- **Function:** Provide energy for movement, growth, reproduction, and body functions.
- **Sources / Ingredients:** Maize, guinea corn, cassava, millet, yam, rice bran, and sweet potato.
- **Deficiency Signs:** Weakness, slow growth, and weight loss.

2. Proteins

- **Function:** Help in growth, repair of body tissues, and production of milk, eggs, and meat.
- **Sources / Ingredients:** Groundnut cake, soybean meal, fish meal, blood meal, beans, cowpea, and pigeon pea.
- **Deficiency Signs:** Poor growth, reduced egg production, and weak muscles.

3. Fats and Oils

- **Function:** Supply concentrated energy and improve the taste and quality of feed; help maintain healthy skin and feathers.
- **Sources / Ingredients:** Palm oil, groundnut oil, coconut oil, cottonseed, and sunflower oil.
- **Deficiency Signs:** Poor body condition, dry skin, and dull feathers.

4. Minerals

- **Function:** Build strong bones and teeth, form blood, and help body functions.
- **Sources / Ingredients:** Bone meal, oyster shell, limestone, salt, and mineral lick blocks.
- **Deficiency Signs:** Weak bones, rickets, and low egg shell quality.

5. Vitamins

- **Function:** Maintain general health, boost disease resistance, and aid growth and reproduction.
- **Sources / Ingredients:** Green leaves, carrots, palm oil, fish oil, sunlight, and vitamin supplements.
- **Deficiency Signs:** Weakness, poor growth, poor appetite, and reduced fertility.

6. Water

- **Function:**
 - Essential for digestion and transportation of nutrients.
 - Helps control body temperature.
 - Maintains life and health.
- **Sources:** Clean drinking water and water in feed ingredients like vegetables and fruits.
- **Deficiency Signs:** Dehydration, loss of appetite, and death if prolonged.

CLASSIFICATION OF FEED INGREDIENTS BY FUNCTION

Feed ingredients can also be grouped based on their **roles in feed formulation**:

Class	Description	Examples
Energy feeds	Provide energy for movement and	Maize, cassava, sorghum, rice

	body functions	bran
Protein feeds	Build and repair body tissues	Groundnut cake, fish meal, soybeans
Mineral supplements	Supply minerals	Bone meal, salt, oyster shell
Vitamin supplements	Supply vitamins	Green vegetables, palm oil, cod liver oil
Roughages	Provide fibre and aid digestion	Grass, hay, silage, crop residues
Additives	Improve feed quality and prevent diseases	Antibiotics, preservatives, antioxidants

FEED FORMULATION

Feed formulation is the process of mixing different feed ingredients in the right proportion to make a balanced diet for animals.

Factors to consider in feed formulation:

1. Type of animal (chicken, goat, pig, cow, etc.)
2. Age of the animal (young, growing, adult, or laying).
3. Purpose of rearing (for meat, milk, or eggs).
4. Availability and cost of feed ingredients.
5. Nutritional value of each ingredient.

Example:

A simple poultry feed may contain:

- 50% maize (energy)
- 20% groundnut cake (protein)

- 10% fish meal (protein)
- 15% wheat bran (energy and fibre)
- 4% bone meal (minerals)
- 1% salt (minerals)

IMPORTANCE OF FEEDING ANIMALS PROPERLY

1. Ensures fast growth and development.
2. Increases production of meat, milk, and eggs.
3. Improves animal health and immunity.
4. Reduces mortality rate.
5. Increases farmer's income and productivity.

SIGNS OF GOOD FEED QUALITY

1. Pleasant smell and free from mould.
2. Dry and not sticky.
3. Uniformly mixed ingredients.
4. No presence of pests or foreign materials.
5. Fresh and properly stored.

SUMMARY

- Feed is any material given to animals for nutrition.
- Feed ingredients are classified into six groups: carbohydrates, proteins, fats and oils, minerals, vitamins, and water.

- Proper feed formulation ensures balanced nutrition.
- The type, age, and purpose of the animal determine the feed combination.
- Good feeding promotes growth, health, and productivity in livestock.

EVALUATION:

1. What is feed?
2. List and explain the six classes of feed nutrients.
3. Mention five examples of carbohydrate feed ingredients.
4. What are protein feeds? Give four examples.
5. State four importance of feeding animals properly.

ASSIGNMENT:

1. Define the term “feed formulation.”
2. List and explain any five factors to consider when formulating animal feed.
3. Identify five signs of good feed quality.
4. Prepare a simple poultry feed formula using at least six ingredients.
5. Explain why water is considered the most important feed ingredient.

SECOND TERM – JSS 1 LIVESTOCK FARMING



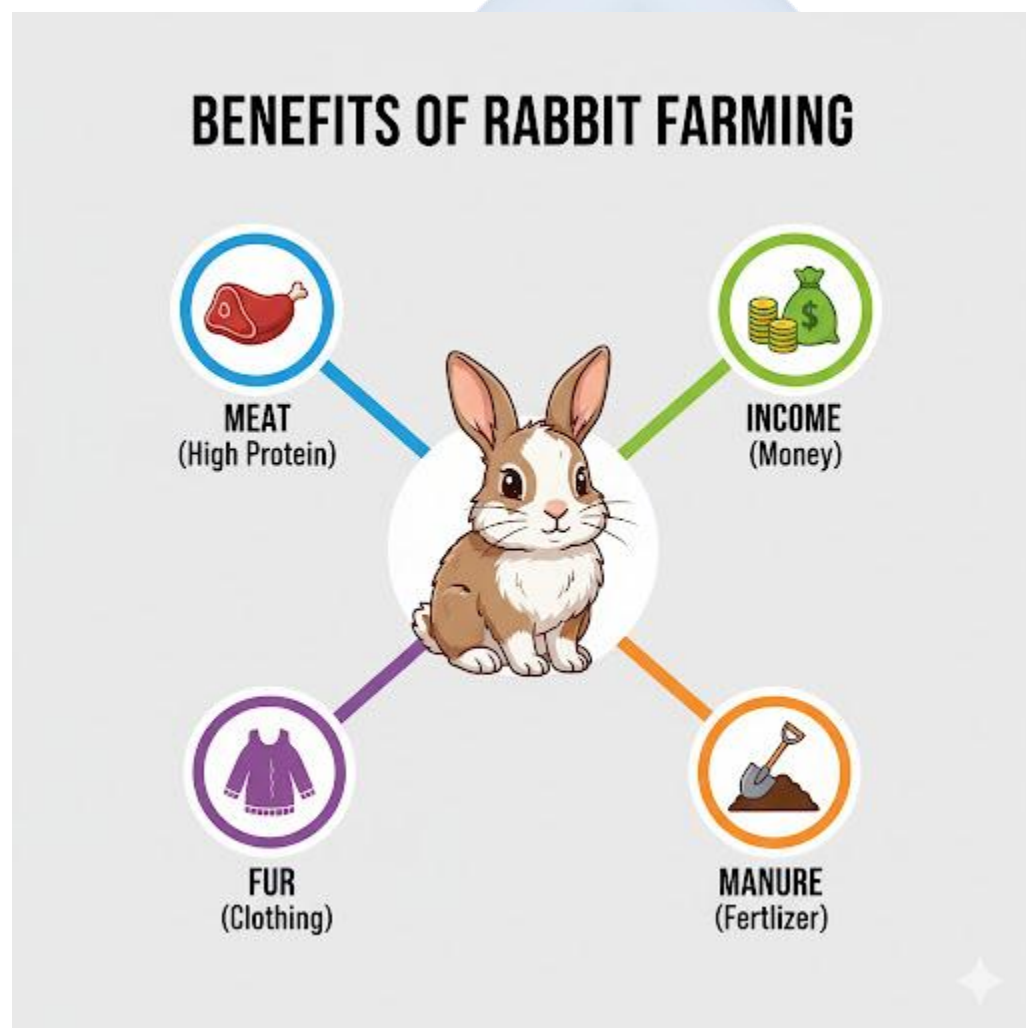
WEEK 1: INTRODUCTION TO RABBIT PRODUCTION

Introduction

Rabbit production is one of the most profitable and fast-growing aspects of livestock farming. Rabbits are small, gentle, and easy-to-raise animals that reproduce quickly and require less space compared to larger farm animals.

They belong to the animal group called **micro-livestock**. Rearing rabbits for food, fur, or as pets is known as **rabbitry**.

Rabbit production provides an excellent source of animal protein and income for farmers. It also helps reduce dependence on beef and poultry meat, making it an important aspect of food security.



MEANING OF RABBIT PRODUCTION

Rabbit production refers to the **breeding, feeding, management, and marketing of rabbits** for economic and domestic purposes.

A place where rabbits are kept is called a **rabbitry**, and the person who raises them is known as a **rabbit farmer** or **rabbit breeder**.

Rabbit production can be done:

- At home (small scale or backyard farming), or
- On a large scale (commercial farming).

IMPORTANCE OF RABBIT PRODUCTION

1. **Source of Meat:**

Rabbit meat is white, tender, tasty, and rich in protein. It is also low in fat and cholesterol, making it healthy.

2. **Source of Income:**

Rabbit farming is profitable because rabbits reproduce quickly and can be sold for meat, fur, or as pets.

3. **Source of Fur and Skin:**

Rabbit fur is soft and used in making hats, gloves, toys, and other clothing materials.

4. **Manure Production:**

Rabbit droppings are an excellent source of organic manure that improves soil fertility.

5. **Educational Purposes:**

Rabbits are used for agricultural science experiments and practicals in schools.

6. **Employment Creation:**

Rabbit farming creates jobs for farmers, feed producers, and traders.

7. **Low Production Cost:**

Rabbits require little space, small amount of feed, and are easy to manage.

8. **Food Security:**

Rabbit meat provides an alternative source of animal protein, reducing pressure on poultry and cattle industries.

9. **Reproduction Efficiency:**

Rabbits multiply rapidly — a single female (doe) can produce up to 6–10 kits (babies) per litter, several times a year.

10. **Environmental Benefits:**

Rabbits do not produce unpleasant odour like some livestock and are suitable for urban and rural environments.

CHARACTERISTICS OF RABBITS

1. Rabbits are **small, gentle, and clean** animals.
2. They are **herbivorous**, feeding mainly on grasses and vegetables.
3. Rabbits are **fast-growing** and mature early (within 4–5 months).
4. They are **prolific breeders**, producing many offspring within a year.
5. They are **nocturnal**, meaning they are more active at night.
6. Rabbits have **long ears** for detecting sound and **strong hind legs** for jumping.
7. They are **social animals** and can be kept in groups if space is adequate.
8. Rabbits have **short gestation periods** (about 30–32 days).
9. They can live for **5 to 8 years** under good management.
10. Rabbits are **sensitive to heat**, so they thrive better in cool environments.

BENEFITS OF RABBIT REARING TO FARMERS AND COMMUNITY

- Provides **healthy white meat** for the family.

- Generates **steady income** through the sale of rabbits and by-products.
- Serves as a **teaching aid** for schools and training institutions.
- Helps in **youth empowerment** through small-scale agribusiness.
- Contributes to **environmental cleanliness** when manure is used as organic fertilizer.

EQUIPMENT AND MATERIALS NEEDED TO START A RABBIT FARM

1. **Hutches:** Wooden or wire cages for housing rabbits.
2. **Feeders and Drinkers:** For feeding and watering.
3. **Nesting Boxes:** For does (female rabbits) to give birth and nurse kits.
4. **Broom and Shovel:** For cleaning droppings.
5. **Water Source:** Clean water supply is essential.
6. **Feed:** Grasses, vegetables, and formulated pellets.
7. **Record Book:** To track breeding, feeding, and health management.

CARE AND MANAGEMENT TIPS FOR BEGINNERS

- Keep the rabbit hutch clean and dry at all times.
- Feed rabbits twice daily with fresh feed and water.
- Protect them from extreme sunlight and rainfall.
- Avoid sudden loud noises or rough handling.
- Separate male and female rabbits except during mating.
- Observe the rabbits daily for signs of sickness.

SUMMARY

- Rabbit production means the rearing of rabbits for food, fur, and income.
- It is simple, profitable, and can be practiced by anyone.
- Rabbits are gentle, fast-growing, and reproduce quickly.
- They provide meat, fur, manure, and income for farmers.
- Good management practices ensure healthy, productive rabbits.

EVALUATION:

1. Define rabbit production.
2. List five characteristics of rabbits.
3. Mention five importance of rabbit farming.
4. State three items needed to start a rabbit farm.
5. Explain two benefits of rabbit rearing to the community.

ASSIGNMENT:

1. What is rabbitry?
2. List and explain five reasons why rabbit production is profitable.
3. Mention five characteristics that make rabbits easy to rear.
4. Identify five equipment or materials needed in rabbit farming and state their uses.
5. Write short notes on the following:
 - (a) Rabbit meat
 - (b) Rabbit manure
 - (c) Rabbit fur

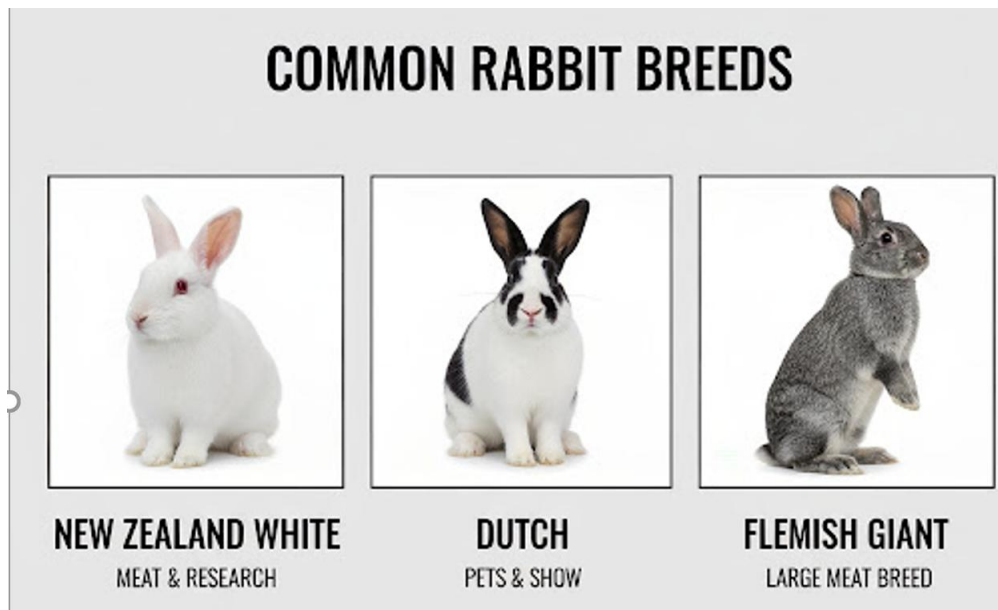
WEEK 2: BREEDS OF RABBIT

Introduction

Rabbits, like other livestock, are divided into **different breeds** based on their size, colour, origin, body shape, and productivity.

Knowing the breeds of rabbits helps farmers to select the right type for meat, fur, or pet production.

Rabbits can be grouped into **local breeds** and **exotic breeds**.



MEANING OF BREED

A **breed** is a group of animals that share similar physical features, origin, and performance characteristics such as size, coat colour, ear length, or rate of growth.

Each rabbit breed has unique qualities that make it suitable for different purposes — some are raised mainly for **meat**, some for **fur**, and others as **pets or laboratory animals**.

CLASSES OF RABBIT BREEDS

Rabbits are generally classified into **two main groups**:

1. **Local Breeds:**

- These are rabbits that have adapted to the local environment and climate conditions.
- They are hardy, can tolerate diseases, and require less attention.
- Their growth rate may be slower, and their body size smaller compared to exotic breeds.
- They are commonly found in rural areas.

2. Exotic Breeds:

- These are foreign breeds introduced from other countries for improved performance.
- They grow faster, produce more meat and fur, and mature earlier.
- However, they require better housing, feeding, and management to perform well.

EXAMPLES OF EXOTIC BREEDS OF RABBITS

Below are some common exotic breeds found in Nigeria and other parts of the world:

1. New Zealand White

- **Origin:** United States of America
- **Colour:** Pure white with red or pink eyes
- **Body Weight:** 4–5 kg when mature
- **Uses:** Meat and fur production
- **Characteristics:**
 - Fast-growing and large-sized breed
 - Docile and easy to handle

- Very prolific (produces many kits per litter)
- Excellent mothering ability

2. Chinchilla Rabbit

- **Origin:** France
- **Colour:** Grey with black-tipped fur (resembles chinchilla rodent)
- **Body Weight:** 3–4.5 kg
- **Uses:** Meat and fur
- **Characteristics:**
 - Fast growth rate and good feed efficiency
 - Produces soft and valuable fur
 - Calm temperament and adapts well to captivity

3. Dutch Rabbit

- **Origin:** Netherlands (Holland)
- **Colour:** Distinct black-and-white or brown-and-white markings
- **Body Weight:** 2–2.5 kg
- **Uses:** Pet and laboratory purposes
- **Characteristics:**
 - Small-sized breed
 - Attractive coat pattern
 - Friendly and suitable for beginners

4. Californian Rabbit

- **Origin:** California, USA
- **Colour:** White body with black or brown markings on nose, ears, and tail
- **Body Weight:** 3.5–4.5 kg
- **Uses:** Mainly for meat production
- **Characteristics:**
 - Fast growth rate and good meat quality
 - Gentle temperament
 - Performs well under good management

5. Angora Rabbit

- **Origin:** Turkey
- **Colour:** White, grey, or cream
- **Body Weight:** 3–4 kg
- **Uses:** Wool and fur production
- **Characteristics:**
 - Produces long, silky wool used for making sweaters and scarves
 - Requires regular grooming and maintenance
 - Slow-growing compared to meat breeds

6. Flemish Giant

- **Origin:** Belgium
- **Colour:** Grey, white, sandy, or black

- **Body Weight:** 6–8 kg (very large breed)
- **Uses:** Meat and exhibition
- **Characteristics:**
 - The largest rabbit breed
 - Calm and friendly in nature
 - Requires large space and more feed

7. Havana Rabbit

- **Origin:** Holland
- **Colour:** Chocolate brown or bluish black
- **Body Weight:** 2.5–3 kg
- **Uses:** Fur and pet purposes
- **Characteristics:**
 - Compact body and glossy coat
 - Good fur texture
 - Friendly and easy to tame

LOCAL BREEDS OF RABBITS (IN NIGERIA)

Local breeds are not as distinct in colour or size as exotic breeds.

Their main features include:

- Small to medium size
- Short brown, grey, or black fur
- Good disease resistance

- Adaptation to local climate
- Slow growth rate
- Low meat yield

Despite their smaller size, local breeds are important because they are hardy and cost less to maintain.

CRITERIA FOR SELECTING A GOOD BREED

When choosing rabbits for production, farmers should consider the following:

1. Purpose of rearing (meat, fur, or pet).
2. Growth rate and body size.
3. Reproductive performance.
4. Feed conversion efficiency (how well they turn feed into meat).
5. Disease resistance and adaptability to the environment.
6. Availability of the breed in the area.

USES OF RABBIT BREEDS

Breed	Primary Use	Special Feature
New Zealand White	Meat and fur	Fast-growing and prolific
Chinchilla	Meat and fur	Attractive grey fur
Dutch	Pet	Beautiful black-and-white coat
Californian	Meat	High-quality meat
Angora	Wool	Long, soft wool

Flemish Giant	Meat and show	Very large size
Local Rabbit	General use	Hardy and disease-resistant

SUMMARY

- Rabbits are classified into **local** and **exotic breeds**.
- Exotic breeds such as New Zealand White, Chinchilla, and Californian grow faster and yield more meat and fur.
- Local breeds are hardy, resistant to diseases, and adapt well to the local climate.
- Breed selection depends on the farmer's purpose and environmental conditions.

EVALUATION:

1. Define a breed.
2. List two classes of rabbit breeds.
3. State five examples of exotic rabbit breeds.
4. Mention four characteristics of the New Zealand White breed.
5. Explain three factors to consider when selecting a rabbit breed.

ASSIGNMENT:

1. Differentiate between local and exotic rabbit breeds.
2. List seven breeds of rabbits and state their primary uses.
3. Describe three reasons why local breeds are still important to Nigerian farmers.
4. Explain why the Angora rabbit requires special care.

5. Write short notes on the following breeds:

(a) Chinchilla

(b) Dutch

(c) Flemish Giant



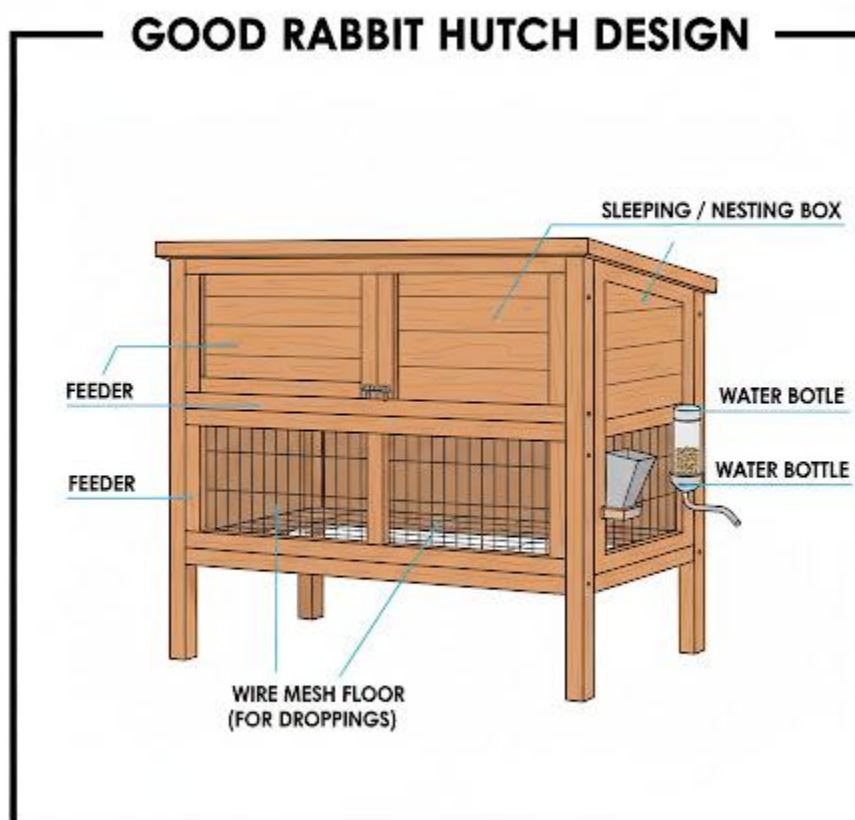
WEEK 3: RABBIT HOUSING

Introduction

Rabbit housing refers to the shelter or structure provided for rabbits to live, feed, reproduce, and stay protected from harsh weather, diseases, and predators.

Good housing is essential in rabbit production because it directly affects their health, growth rate, reproduction, and productivity.

A well-constructed house makes management easier, reduces mortality, and ensures the rabbits are safe, comfortable, and productive.



REASONS FOR PROVIDING HOUSING FOR RABBITS

1. **Protection from predators** – such as dogs, snakes, cats, and hawks.

2. **Protection from harsh weather conditions** – like heavy rain, sun, and cold.
3. **For better management** – makes feeding, cleaning, and breeding easier.
4. **Disease prevention** – good housing reduces contact with waste and parasites.
5. **Comfort and growth** – clean and spacious housing allows rabbits to grow and reproduce well.
6. **Record keeping and control** – proper housing allows identification and monitoring of each rabbit.

TYPES OF RABBIT HOUSING

Rabbits can be kept in various housing systems depending on the number of rabbits, available space, and resources of the farmer. The three major housing systems are:

1. Hutch System

- A **hutch** is a small, enclosed cage made of wood, wire mesh, and metal sheets.
- It can be used for keeping a single rabbit or a small group.
- It may be **kept indoors or outdoors** depending on the climate.
- Each hutch usually contains separate sections for feeding, resting, and breeding.

Advantages:

- Easy to clean and manage.
- Prevents fighting among rabbits.
- Helps to monitor health and feeding easily.

Disadvantages:

- Not suitable for large-scale rabbit farming.
- Construction requires time and materials.

2. Deep Litter System

- In this system, rabbits are kept in a room or pen where the floor is covered with dry litter materials such as wood shavings or straw.
- It allows several rabbits to move freely in the same space.

Advantages:

- Suitable for rearing many rabbits together.
- Cheaper to set up on a large scale.

Disadvantages:

- Difficult to monitor individual rabbits.
- Diseases can spread quickly due to droppings.
- Requires frequent cleaning and change of litter.

3. Cage System

- This involves using **metallic or wire cages** arranged in tiers (single, double, or triple).
- Commonly used in commercial rabbit farms.

Advantages:

- Efficient use of space.
- Easy waste collection through droppings trays.
- Ideal for commercial production.

Disadvantages:

- Expensive to install.
- May cause sore feet if wire mesh flooring is not properly constructed.

COMPONENTS OF A RABBIT HOUSE (HUTCH)

A good rabbit hutch or house should contain the following parts:

1. **Feeding Trough or Feeder:** For supplying feed.
2. **Drinker or Water Trough:** For providing clean water at all times.
3. **Resting Area:** A sheltered or covered part where rabbits sleep and rest.
4. **Nest Box:** For pregnant does (female rabbits) to give birth and nurse their kits.
5. **Ventilation Openings:** For fresh air and to prevent odour.
6. **Dropping Tray:** Placed below cages to collect waste and ease cleaning.

REQUIREMENTS OF A GOOD RABBIT HOUSE

1. **Dry and Clean Environment:** Moisture encourages diseases and odour.
2. **Proper Ventilation:** To allow air circulation and prevent suffocation.
3. **Protection from Weather:** The house must shield rabbits from direct sunlight, wind, and rain.
4. **Ease of Cleaning:** The structure should be easy to wash or disinfect.
5. **Adequate Space:** Each rabbit should have enough room to move freely.
6. **Security:** The house should prevent entry of predators.
7. **Proper Drainage:** To remove urine and waste easily.

MATERIALS USED FOR CONSTRUCTING A RABBIT HOUSE

- **Wood:** For frame and partitions.
- **Wire mesh:** For doors, sides, and flooring (to allow ventilation and waste removal).

- **Nails and hinges:** For joining and fixing doors.
- **Iron sheets or thatch:** For roofing to protect from sun and rain.
- **Cement or gravel:** For solid flooring (in large farms).

FACTORS TO CONSIDER WHEN CONSTRUCTING A RABBIT HOUSE

1. **Location:** Should be in a quiet, well-drained area away from noise and predators.
2. **Orientation:** The front should face east–west to minimize direct sunlight.
3. **Number of Rabbits:** The size depends on the number and type of rabbits to be kept.
4. **Ventilation and Drainage:** To ensure comfort and cleanliness.
5. **Ease of Access:** For feeding, cleaning, and inspection.
6. **Type of Production:** Whether it is for breeding, fattening, or as pets.

SIZES AND SPACE REQUIREMENTS

- **Breeding Buck (male):** 0.75 m × 0.75 m × 0.45 m
- **Doe with Litter:** 1.0 m × 0.75 m × 0.45 m
- **Growing Rabbits:** 0.6 m × 0.6 m × 0.45 m

Each rabbit should have enough room to move, stretch, and feed without stress.

IMPORTANCE OF GOOD RABBIT HOUSING

- Enhances **health** and reduces disease.
- Encourages **reproduction and growth**.
- Simplifies **feeding, cleaning, and breeding**.
- Reduces **mortality**.

- Promotes **efficient record keeping** and management.

SUMMARY

- Rabbit housing provides shelter, safety, and comfort.
- Main housing systems: **Hutch, Deep Litter, and Cage Systems.**
- A good rabbit house must be **dry, clean, well-ventilated, spacious, and secure.**
- Proper materials like **wood, wire mesh, and roofing sheets** are essential.
- The housing system chosen should depend on the farmer's resources and the number of rabbits.

EVALUATION:

1. Define rabbit housing.
2. State three reasons why rabbits need housing.
3. List and explain the three main types of rabbit housing.
4. Mention five components of a rabbit house.
5. State four materials used in constructing a rabbit hutch.

ASSIGNMENT:

1. Describe five requirements of a good rabbit house.
2. Explain three advantages of the cage system over the deep litter system.
3. List six factors to consider before constructing a rabbit house.
4. Differentiate between a hutch and a cage system.
5. Draw and label a simple diagram of a rabbit hutch, showing its main parts.

WEEK 4: RABBIT FEEDS

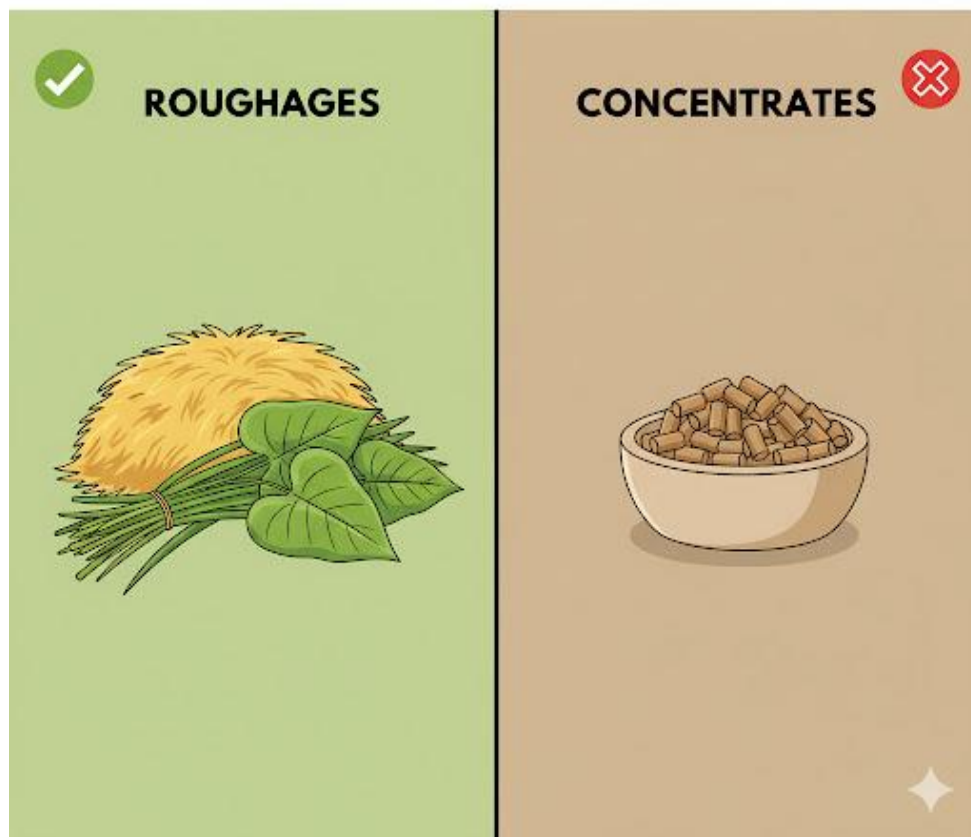
Introduction

Feeding is one of the most important aspects of rabbit production. The type and quality of feed given to rabbits greatly influence their growth rate, health, reproduction, and meat quality.

Rabbits are herbivores — animals that eat only plant materials — and they require a balanced diet to grow well, reproduce efficiently, and remain healthy.

A good feeding programme must provide all the nutrients that rabbits need for body maintenance, growth, reproduction, and milk production.

A RABBIT'S DIET



MEANING OF FEED

A **feed** is any substance (solid or liquid) that can be eaten or drunk by an animal to supply nutrients necessary for its survival, growth, and reproduction.

For rabbits, feed includes **fresh grasses, vegetables, hay, concentrates, and clean drinking water.**

IMPORTANCE OF FEEDING RABBITS

1. Provides nutrients for body growth and repair.
2. Maintains good health and prevents diseases caused by malnutrition.
3. Enhances reproduction, pregnancy, and milk production in does (female rabbits).
4. Improves meat and fur quality.
5. Increases weight gain and overall productivity.
6. Reduces mortality and stress in rabbits.

TYPES OF RABBIT FEEDS

Rabbit feeds are grouped into two major types:

1. Concentrates

- These are **energy-rich and nutrient-dense** feeds.
- They contain little fibre and are easily digestible.
- Examples: commercial rabbit pellets, maize, wheat bran, groundnut cake, soybean meal, fish meal, and bone meal.

Advantages of Concentrates:

- Encourage fast growth and weight gain.
- Supply essential nutrients (proteins, carbohydrates, minerals, vitamins).
- Reduce feeding space and waste.

Disadvantages:

- More expensive than grasses.
- Overfeeding can lead to obesity or digestive problems.

2. Roughages

- These are **fibrous, bulky feeds** that help in digestion and keep the rabbit's teeth in good condition.
- They form the **main part of a rabbit's diet**.
- Examples: fresh grasses, legumes, hay, leaves, vegetables, and root crops.

Examples of Roughages Commonly Fed to Rabbits:

- Elephant grass
- Guinea grass
- Tridax leaves
- Sweet potato leaves
- Pawpaw leaves
- Moringa leaves
- Carrot tops
- Cassava leaves (sun-dried before feeding)
- Maize and groundnut haulms
- Cabbage and spinach leaves

Importance of Roughages:

- Help digestion through fibre content.
- Prevent overgrowth of rabbit teeth.
- Cheap and easily available.

- Provide vitamins and minerals.

BALANCED RABBIT FEED

A **balanced rabbit feed** contains all six essential classes of nutrients in the correct proportion:

Nutrient	Function in Rabbits	Examples of Sources
Carbohydrates	Provide energy	Maize, wheat bran, cassava peel
Proteins	Build body tissues, aid growth and reproduction	Groundnut cake, soybean meal, fish meal
Fats and Oils	Provide energy and improve fur	Palm oil, groundnut oil
Minerals	Strengthen bones and teeth	Bone meal, oyster shell
Vitamins	Support health and reproduction	Green vegetables, carrots
Water	Aid digestion and regulate temperature	Clean, fresh water daily

FEEDING MANAGEMENT PRACTICES

1. **Regular Feeding Schedule:** Feed rabbits at the same time every morning and evening.
2. **Provide Clean Water:** Always ensure constant access to clean water.
3. **Avoid Sudden Feed Change:** Introduce new feeds gradually to prevent stomach upset.
4. **Use Proper Feeders:** To prevent feed wastage and contamination.
5. **Feed According to Age:**
 - Kits (young rabbits): More milk and soft feeds.
 - Growers: Balanced feed for rapid growth.
 - Adults: Maintenance and reproduction feed.
6. **Remove Leftovers Daily:** To maintain cleanliness and prevent mould growth.

7. **Provide Salt Licks:** To supply extra minerals.
8. **Include Fibre (hay):** Always ensure hay is available to aid digestion.

FEEDING SCHEDULES FOR RABBITS

Rabbit Category	Type of Feed	Feeding Frequency
Kits (1–4 weeks)	Mother's milk	Suckling until weaning
Growers (5–10 weeks)	Concentrates + Roughages	Twice daily
Breeding Bucks (Males)	Concentrates + Greens	Once or twice daily
Does (Pregnant or Nursing)	Concentrates + Greens + Hay	Twice daily
Weaned Rabbits	Balanced feed and water	Twice daily

WATER MANAGEMENT IN RABBITRY

Water is as important as feed. Rabbits require **fresh, clean water daily** for digestion, temperature regulation, and milk production.

- Use **clean drinkers or bottles** to supply water.
- Wash the containers daily.
- Never allow water to stay dirty or stagnant.

COMMON FEEDING PROBLEMS

1. **Overfeeding:** Causes obesity and low fertility.
2. **Underfeeding:** Leads to stunted growth and poor reproduction.
3. **Dirty Feed or Water:** Causes diarrhoea and diseases.
4. **Unbalanced Feed:** Results in poor fur quality and weak bones.

5. **Lack of Fibre:** Causes digestive problems and teeth overgrowth.

SUMMARY

- Rabbits are herbivores and feed mainly on plants.
- They require both **concentrates** (rich feeds) and **roughages** (fibrous feeds).
- Feeding must be regular, clean, and well-balanced.
- Fresh water should always be available.
- Proper feeding ensures healthy growth, good reproduction, and quality meat.

EVALUATION:

1. Define rabbit feed.
2. List the two main types of rabbit feeds.
3. Give five examples of roughages fed to rabbits.
4. Mention four examples of concentrate feeds.
5. State four feeding management practices in rabbit farming.

ASSIGNMENT:

1. Explain the difference between concentrates and roughages.
2. Write out six classes of nutrients needed in rabbit feed and give one source of each.
3. Mention four effects of poor feeding in rabbits.
4. State five ways of ensuring good water management in rabbitry.
5. Design a **one-day feeding schedule** for a mature doe (female rabbit) showing time, type of feed, and quantity.

WEEK 6: HEALTH MANAGEMENT IN RABBITS

Introduction

Health management in rabbits refers to all the practices carried out to ensure that rabbits remain healthy, free from diseases, and capable of producing efficiently.

Good health is essential for profitable rabbit farming. Rabbits are delicate animals that require clean surroundings, balanced feeding, and proper handling to stay healthy.

Health management involves disease prevention, treatment, proper housing, and observation of rabbits daily.

RABBIT HEALTH



IMPORTANCE OF HEALTH MANAGEMENT IN RABBITRY

1. **To prevent disease outbreak** among rabbits.

2. **To promote growth and productivity** (good meat and fur quality).
3. **To reduce mortality and losses** in the rabbit farm.
4. **To ensure efficient reproduction** and milk production in does.
5. **To reduce cost of treatment** by preventing sickness before it occurs.
6. **To promote the welfare and comfort** of rabbits.

COMMON RABBIT DISEASES, SYMPTOMS, AND CONTROL MEASURES

Below are some of the common diseases and health problems that affect rabbits, their causes, symptoms, and preventive measures.

1. Coccidiosis

- **Cause:** A protozoan parasite (Coccidia) found in contaminated food, water, or litter.
- **Symptoms:**
 - Loss of appetite
 - Diarrhoea (sometimes bloody)
 - Weight loss
 - Dull coat and weakness
- **Prevention/Control:**
 - Keep hutches clean and dry.
 - Avoid overcrowding.
 - Regularly disinfect the cages and utensils.
 - Add coccidiostat drugs (e.g. Sulfaquinoxaline) to feed or water under a vet's advice.

2. Pasteurellosis (Snuffles)

- **Cause:** A bacterial infection (*Pasteurella multocida*).
- **Symptoms:**
 - Sneezing and nasal discharge
 - Difficulty breathing
 - Watery eyes
 - Loss of appetite
- **Prevention/Control:**
 - Keep environment clean and dust-free.
 - Avoid sudden temperature changes.
 - Isolate and treat infected rabbits with antibiotics (as prescribed by a vet).

3. Ear Mange

- **Cause:** Tiny mites that burrow into the skin of the rabbit's ear.
- **Symptoms:**
 - Scratching and shaking of ears
 - Thick brown crust inside ears
 - Hair loss around ears
- **Prevention/Control:**
 - Regularly check ears for mites.
 - Apply approved anti-parasitic oil or ivermectin.
 - Maintain proper hygiene and sanitation.

4. Myxomatosis

- **Cause:** A virus transmitted by mosquitoes, fleas, or direct contact with infected rabbits.
- **Symptoms:**
 - Swelling of the eyes, lips, and genitals
 - Discharge from eyes and nose
 - Loss of appetite and lethargy
- **Prevention/Control:**
 - Vaccinate rabbits against myxomatosis.
 - Control insects and pests around the farm.
 - Quarantine newly bought rabbits before mixing with others.

5. Mastitis

- **Cause:** Bacterial infection of the mammary glands in nursing does.
- **Symptoms:**
 - Swollen and painful breasts
 - Redness and heat around the mammary area
 - Refusal to feed kits
- **Prevention/Control:**
 - Keep nest boxes clean and dry.
 - Avoid injuries to the mammary glands.
 - Treat affected rabbits with antibiotics as prescribed by a vet.

6. Diarrhoea

- **Cause:** Poor-quality feed, sudden feed change, or dirty water.
- **Symptoms:**
 - Frequent watery stool
 - Weakness and loss of appetite
- **Prevention/Control:**
 - Provide clean feed and water.
 - Avoid mouldy or spoiled feed.
 - Give oral rehydration or antibiotics if necessary.

PARASITES THAT AFFECT RABBITS

Rabbits are also attacked by **external and internal parasites**.

1. External Parasites

These live **on the body surface** of the rabbit.

Examples: Lice, fleas, mites, and ticks.

Effects:

- Cause irritation and scratching.
- Lead to hair loss and skin wounds.
- May transmit diseases.

Control Measures:

- Regular grooming and cleaning.
- Use of anti-parasitic sprays or powders.
- Disinfection of hutches and equipment.

2. Internal Parasites

These live **inside the rabbit's body**, especially in the intestines.

Examples: Roundworms and tapeworms.

Effects:

- Reduced feed intake and weight loss.
- Diarrhoea and poor growth.
- Weakness and death in severe cases.

Control Measures:

- Deworm rabbits regularly.
- Keep feed and water clean.
- Maintain good hygiene in housing areas.

GENERAL PREVENTIVE MEASURES IN RABBITRY

1. **Maintain cleanliness** – wash hutches, feeders, and drinkers daily.
2. **Provide clean feed and water** at all times.
3. **Avoid overcrowding** to prevent disease spread.
4. **Disinfect equipment** regularly using approved disinfectants.
5. **Isolate sick animals** immediately from healthy ones.
6. **Practice quarantine** for newly introduced rabbits.
7. **Control pests and insects** around the farm.
8. **Consult a veterinarian** regularly for check-ups and vaccination.

SYMPTOMS OF A HEALTHY RABBIT

- Bright and alert eyes
- Smooth, clean fur
- Active movement
- Normal feeding and drinking
- Clean nose and ears
- Normal droppings (firm and dry pellets)

SYMPTOMS OF A SICK RABBIT

- Dull or droopy eyes
- Loss of appetite
- Rough or dirty coat
- Isolation from others
- Diarrhoea or constipation
- Nasal discharge or sneezing
- Lethargy and unusual behaviour

SUMMARY

- Health management ensures rabbits remain strong, disease-free, and productive.
- Common diseases include **Coccidiosis, Pasteurellosis, Ear Mange, Myxomatosis, Mastitis, and Diarrhoea.**
- Diseases can be controlled through **cleanliness, vaccination, isolation, and good feeding.**
- Healthy rabbits show **bright eyes, clean fur, and active behaviour.**

EVALUATION:

1. Define health management in rabbits.
2. List five importance of rabbit health management.
3. Name four common diseases of rabbits and state one symptom of each.
4. Mention two internal and two external parasites of rabbits.
5. State four preventive measures against rabbit diseases.

ASSIGNMENT:

1. Explain briefly three causes of poor health in rabbits.
2. Write four symptoms of a healthy rabbit and four symptoms of a sick rabbit.
3. Mention five ways of controlling external parasites in rabbit farms.
4. Describe three ways to prevent Coccidiosis.
5. Draw a table showing **disease, cause, symptom, and control** for three rabbit diseases.

WEEK 8: REPRODUCTION IN RABBITS

Meaning of Reproduction in Rabbits

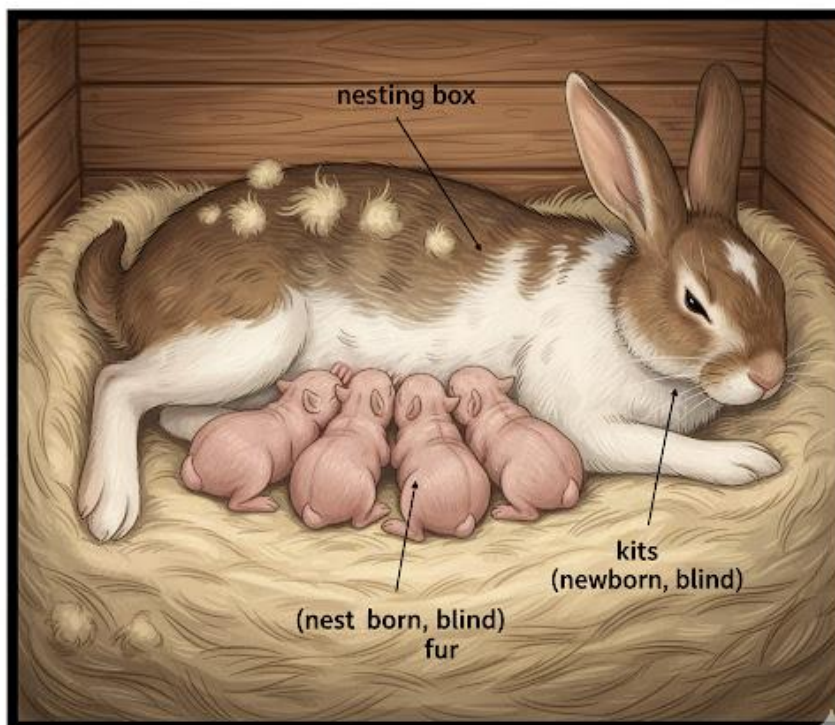
Reproduction in rabbits is the natural biological process through which rabbits produce young ones. It involves mating between a male rabbit (buck) and a female rabbit (doe), fertilization, pregnancy (gestation), kindling (giving birth), and caring for the young until they are weaned.

Efficient reproduction management ensures steady production of rabbits for meat, fur, or breeding purposes.

The Reproductive Organs of Rabbits

- **Male (Buck):** Testes, penis, and accessory glands — produce and release sperm.
- **Female (Doe):** Ovaries, oviducts, uterus, and vagina — produce eggs and provide a site for fertilization and development of the young.

RABBIT REPRODUCTION



The Mating Process

- Mating should always take place **in the buck's cage** because male rabbits perform better in their own territory.
- The doe is placed in the buck's cage and mating usually occurs within a few minutes.
- After mating, the doe should be returned to her cage immediately to prevent fighting.
- A second mating can be allowed after a few minutes for better conception.

Signs of Heat (When the Doe is Ready for Mating):

- Restlessness and rubbing her chin on objects.
- Swollen and reddish vulva.
- Frequent mounting of other rabbits.
- Loss of appetite.

Gestation Period

- Gestation is the period between fertilization and kindling (birth).
- It lasts about **28–31 days** in rabbits.
- During gestation, the doe should be given nutritious feed and kept in a quiet environment.
- Provide a **nesting box** lined with soft hay a few days before kindling.

Kindling (Giving Birth)

- Kindling is the act of a doe giving birth to her kits (baby rabbits).
- It usually happens quietly at night or early morning.
- The doe pulls fur from her body to line the nest for warmth.

- Average litter size ranges from **6–10 kits**, depending on the breed.
- Avoid disturbing the doe for the first few days after kindling to prevent stress.

Weaning

- Weaning means separating the young rabbits from the mother.
- It usually occurs between **4–6 weeks of age**.
- The young ones should be moved to a separate cage and given soft feeds like green vegetables and pellets.
- Proper care during weaning helps them adapt quickly and reduces mortality.

Care of the Young Rabbits (Kits)

- Keep the nest warm and dry.
- Ensure they suckle well for the first few weeks.
- Provide clean water and fresh feed after weaning.
- Watch for signs of sickness or weakness.

EVALUATION

1. What is reproduction in rabbits?
2. State two functions of the male and female reproductive organs.
3. Mention three signs that show a doe is ready for mating.
4. What is the gestation period of rabbits?
5. Define kindling and explain two care practices after kindling.

ASSIGNMENT

1. Explain the process of mating in rabbits.
2. Describe how to care for a pregnant doe.
3. State the duration and importance of the gestation period.
4. What is weaning? At what age should kits be weaned?
5. Mention three reasons why a farmer should not disturb a doe during kindling.



WEEK 9: RECORD KEEPING IN RABBITRY

Meaning of Record Keeping

Record keeping in rabbit production means the systematic documentation of important information about the animals, their feed, breeding, health, and sales. Proper record keeping helps a farmer track the progress of the rabbitry, monitor productivity, and make informed management and financial decisions.

Without proper records, a rabbit farmer may find it difficult to know whether the business is making profit or loss, or to identify the cause of poor performance.

Doe ID	Mating Date	Kindling Date	No. Born	No. Weaned
F-003	2023-10-15	2023-11-15	8	6

Importance of Record Keeping in Rabbit Production

1. Helps to monitor performance:

- Records show how each rabbit grows, reproduces, and produces meat or fur.

2. Assists in planning and decision-making:

- Farmers can determine which breeds perform best and make future plans accordingly.

3. Helps in disease control:

- Health records make it easier to identify recurring diseases and take preventive measures.

4. **Improves breeding efficiency:**

- Breeding records help in avoiding inbreeding and selecting the best animals for mating.

5. **Assists in financial management:**

- By comparing expenditure and income records, farmers can calculate profit or loss.

6. **Provides reliable data for research or loan applications:**

- Accurate records can be used to apply for government support, loans, or training programmes.

Types of Records Kept in Rabbitry

1. **Breeding Record**

- This record contains information about mating, gestation, kindling, and weaning.
- It includes dates of mating, name or number of the buck and doe, number of kits born and weaned, and remarks on success or failure.

Example of Breeding Record Table:

Doe ID	Buck ID	Date Mated	Kindling Date	No. Born	No. Weaned	Remarks
D001	B002	12/01/2025	10/02/2025	8	7	Successful

2. **Feeding Record**

- Shows the type and quantity of feed given daily or weekly.
- Helps to monitor feed consumption and control feeding cost.

Example:

Date	Feed Type	Quantity (kg)	Remarks
05/02/2025	Concentrate	2.5	Normal feeding

3. Health Record

- Contains information on disease occurrence, treatment, vaccination, and mortality.
- Helps track the health history of each rabbit.

Example:

Date	Rabbit ID	Disease	Treatment	Vet Name	Remarks
08/02/2025	D004	Coccidiosis	Amprolium	Mr. Ayo	Recovered

4. Sales Record

- Contains information on rabbits sold, prices, and total income.
- Helps in determining the profitability of the rabbit business.

Example:

Date	Buyer	No. Sold	Price per Rabbit (₦)	Total (₦)
12/03/2025	Mrs. Edet	5	6,000	30,000

5. Expenditure Record

- Records all money spent on feed, medication, equipment, and maintenance.
- Used to calculate profit or loss at the end of the production cycle.

Example:

Date	Item Purchased	Cost (₦)	Remarks
03/03/2025	Feed	12,000	Monthly supply

Qualities of a Good Record

- Must be **accurate** and **clear**.
- Must be **up-to-date**.
- Should be **neatly arranged and easy to read**.
- Should be **stored safely** to prevent loss or damage.
- Must be **based on real facts** and not guesses.

EVALUATION

1. What is record keeping in rabbit production?
2. Mention four reasons why record keeping is important in rabbit farming.
3. List and explain three types of records kept in a rabbitry.
4. State three qualities of a good record.
5. Give two examples of information that can be found in a breeding record.

ASSIGNMENT

1. Define record keeping and explain its importance in rabbit production.
2. List five different types of records a rabbit farmer should keep.
3. Describe the information that should be recorded in a feeding record.
4. Explain how record keeping helps in disease control.
5. Design a simple table for keeping health records in a rabbit farm.



WEEK 10: HANDLING AND MANAGEMENT PRACTICES IN RABBITRY

Meaning of Handling and Management Practices

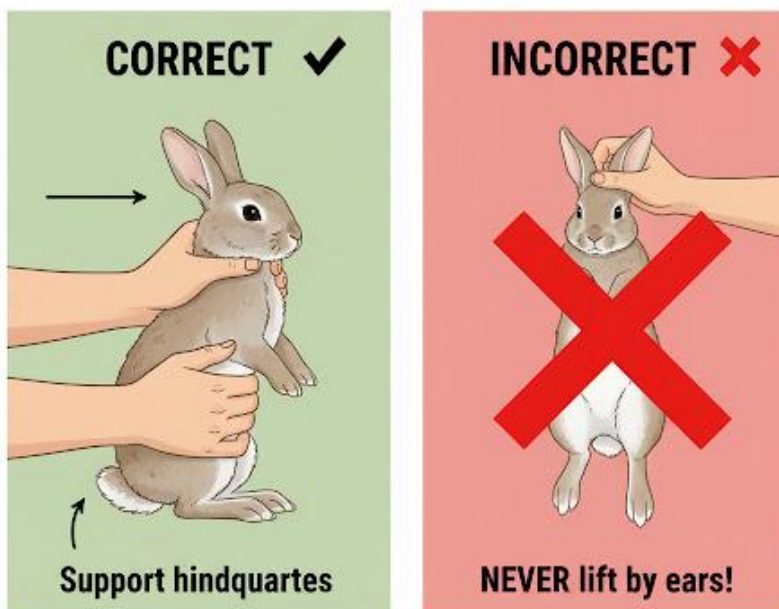
Handling and management practices in rabbit production refer to all the routine activities carried out to ensure that rabbits remain healthy, comfortable, and productive. Proper handling helps prevent injuries and stress, while good management ensures that rabbits grow well and reproduce efficiently.

HANDLING OF RABBITS

Meaning of Handling

Handling means the way a farmer holds, lifts, or moves rabbits from one place to another. Because rabbits are delicate animals, careless handling can cause injuries, broken bones, or even death. Therefore, every rabbit farmer must learn the correct way to handle rabbits.

HOW TO HANDLE A RABBIT



Correct Methods of Handling Rabbits

1. **Hold firmly but gently:** Avoid squeezing or frightening the rabbit.
2. **Lift by supporting the hindquarters:**
 - One hand should hold the loose skin behind the ears (scruff) while the other supports the hind legs.
3. **Avoid lifting by the ears:** This causes severe pain and may damage the ear cartilage.
4. **Keep rabbits close to your body** when carrying them to prevent struggling or falling.
5. **Use a carrier or basket** when moving rabbits over a long distance.

Reasons for Proper Handling

- Prevents injury or death of the animal.
- Reduces stress and improves feed intake.
- Makes examination, treatment, and weighing easier.
- Promotes trust and calmness in the rabbits.

MANAGEMENT PRACTICES IN RABBITRY

1. Grooming

- Grooming means cleaning and maintaining the rabbit's body and fur.
- It involves brushing the fur, trimming long nails, and checking for parasites.
- Grooming prevents hair matting, improves appearance, and enhances comfort.

2. Sanitation (Cleaning and Disinfection)

- Sanitation involves cleaning hutches, feeders, and drinkers daily.
- Remove droppings and leftover feed regularly.
- Disinfect the rabbit house weekly using mild disinfectants to kill germs.

- Good sanitation prevents diseases and keeps the rabbitry odor-free.

3. Waste Disposal

- Rabbit droppings and bedding waste should be collected daily.
- Dispose of waste properly by composting or burying it away from the hutch.
- Proper waste management prevents flies, odor, and disease outbreak.

4. Feeding and Watering

- Provide clean and fresh feed and water at all times.
- Feed should contain all nutrients needed for growth and reproduction.
- Clean feeders and drinkers daily to prevent contamination.

5. Housing Management

- Keep hutches dry, well-ventilated, and safe from predators.
- Repair damaged cages and replace worn-out materials promptly.
- Ensure adequate space for each rabbit to move freely.

6. Health Care

- Observe rabbits daily for signs of illness such as dullness, loss of appetite, or discharge from eyes and nose.
- Isolate sick rabbits immediately for treatment.
- Deworm and vaccinate as recommended.

7. Breeding Management

- Mate rabbits at the appropriate age (doe: 5–6 months; buck: 6–7 months).
- Avoid inbreeding by keeping breeding records.
- Provide nest boxes for pregnant does before kindling.

IMPORTANCE OF GOOD MANAGEMENT PRACTICES

- Promotes rapid growth and weight gain.
- Increases reproduction and productivity.
- Prevents spread of diseases.
- Reduces death rate and loss of investment.
- Enhances quality of meat, fur, and other rabbit products.

EVALUATION

1. What is meant by handling in rabbit farming?
2. Mention three correct ways of handling rabbits.
3. Explain three management practices in rabbit production.
4. State three reasons for grooming rabbits.
5. List four importance of good management practices in rabbit production.

ASSIGNMENT

1. Describe the correct method of handling rabbits.
2. Explain the importance of sanitation in rabbit management.
3. List five management practices every rabbit farmer should perform regularly.
4. Discuss how improper handling can affect rabbit production.
5. Outline two ways to dispose of rabbit waste safely.

WEEK 11: REVISION AND PRACTICAL DEMONSTRATION — FEEDING, CLEANING, AND MAINTENANCE

This week's focus is on reinforcing knowledge and developing practical skills in daily rabbit management.

PRACTICAL AREAS OF FOCUS

1. Feeding Rabbits

Feeding is one of the most important management practices in rabbit production. It ensures proper growth, reproduction, and overall good health.

Feeds Commonly Used for Rabbits:

- **Concentrates:** Commercial pellets, maize bran, wheat offal, groundnut cake.
- **Roughages (Green Feeds):** Grasses, legumes, vegetables, and leaves like pawpaw leaves, waterleaf, and guinea grass.
- **Water:** Fresh, clean water must be provided at all times.

Feeding Procedures:

1. Clean all feeders before use.
2. Provide the right quantity of feed according to age and size.
3. Feed twice daily — morning and evening.
4. Remove leftover feed to avoid spoilage.
5. Observe each rabbit while feeding to ensure all are eating properly.

Signs of Proper Feeding:

- Active movement and playfulness.
- Smooth and shiny fur.
- Steady growth and weight gain.

- Regular droppings (firm and dry).

2. Cleaning and Sanitation

A clean environment is essential for keeping rabbits healthy and preventing diseases.

Cleaning Procedures:

- Remove droppings and old bedding materials daily.
- Wash feeders and drinkers with clean water and mild soap.
- Disinfect hutches weekly using an approved disinfectant.
- Keep the surroundings dry and free from stagnant water.

Importance of Cleaning and Sanitation:

- Prevents disease outbreaks and foul odour.
- Reduces fly and parasite infestation.
- Improves air quality in the rabbit house.
- Enhances rabbit comfort and performance.

3. Maintenance of Rabbitry

Maintenance involves all the activities carried out to keep the rabbit house, equipment, and animals in good condition.

Maintenance Activities Include:

- Repairing broken or damaged hutches.
- Replacing worn-out feeders and drinkers.
- Checking for leaks or sharp objects that can harm rabbits.
- Ensuring proper ventilation and light in the hutch.

- Regularly inspecting rabbits for signs of illness or injury.

Benefits of Regular Maintenance:

- Extends the lifespan of equipment and hutches.
- Promotes continuous production.
- Reduces cost of replacement or loss.
- Ensures safety and comfort for both rabbits and handlers.

4. Demonstration Procedures

During practical sessions, students should:

1. Observe how feed is measured and given to rabbits.
2. Participate in cleaning hutches and disinfecting the environment.
3. Assist in identifying healthy and unhealthy rabbits.
4. Learn how to hold and transfer rabbits properly.
5. Record feeding and cleaning activities in a farm notebook.

EVALUATION

1. Mention three types of feed given to rabbits.
2. Outline the daily routine for feeding rabbits.
3. State four cleaning activities carried out in a rabbit farm.
4. Mention three maintenance activities in rabbit management.
5. List three signs that show rabbits are properly fed and healthy.

ASSIGNMENT

1. Describe step-by-step how to feed and water rabbits daily.
2. Explain the importance of regular cleaning in rabbit management.
3. List five maintenance activities every rabbit farmer should perform.
4. Describe the correct procedure for disinfecting a rabbit hutch.
5. Prepare a one-week record sheet for feeding and cleaning practices in your school rabbitry.



THIRD TERM – JSS 1 LIVESTOCK FARMING

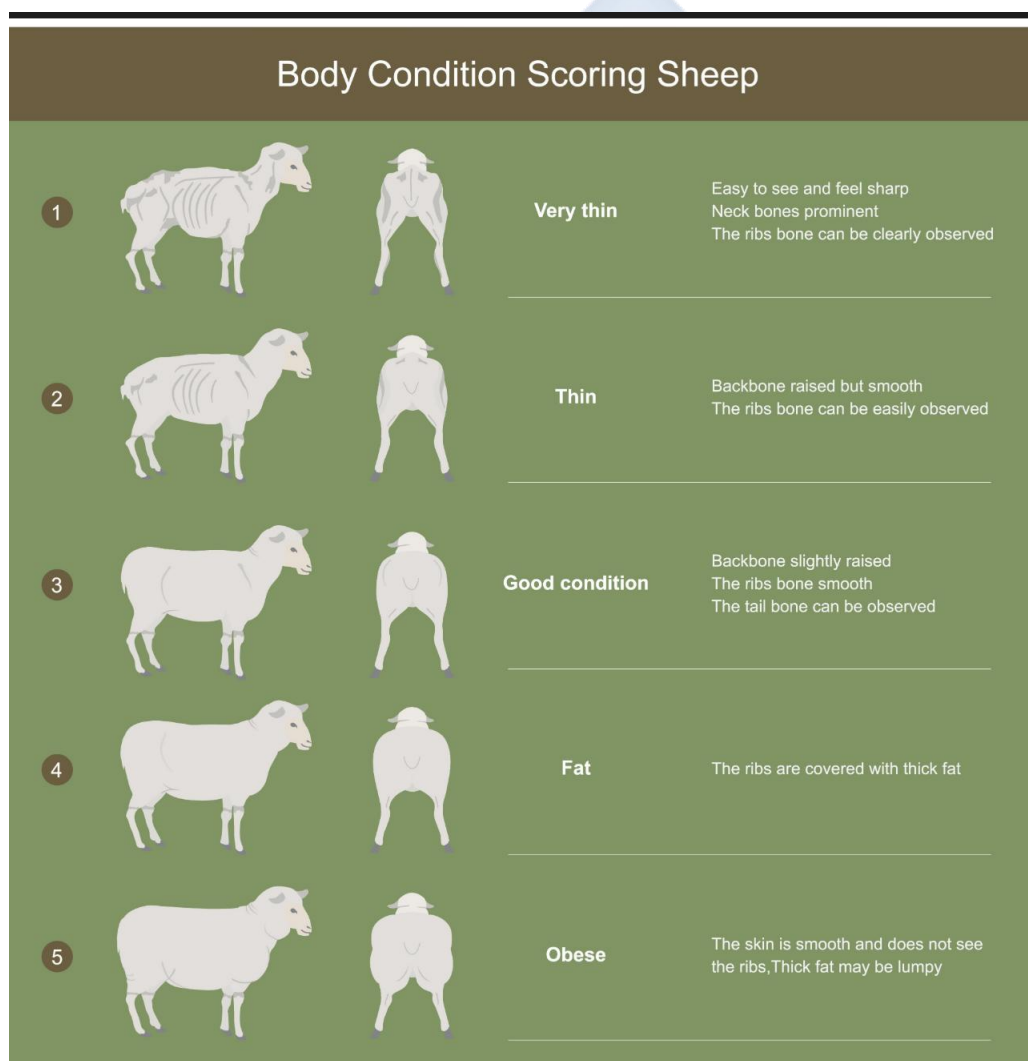


WEEK 1: CONCEPT OF SHEEP AND GOAT PRODUCTION

Meaning of Sheep and Goat Production

Sheep and goat production refers to the rearing, breeding, and management of sheep and goats for the purpose of obtaining meat, milk, hides, skin, wool, manure, and income. These animals are among the most important small ruminants kept by farmers across Nigeria and other parts of Africa.

They are hardy animals that can survive in different environmental conditions and require relatively low cost of production, making them ideal for both small- and large-scale farming.



Importance of Sheep and Goat Production

1. Source of Meat:

- Both animals produce highly nutritious meat rich in protein.
- Goat meat is commonly known as *chevon*, and sheep meat as *mutton*.

2. Source of Milk:

- Goat milk is a valuable source of food and can be processed into cheese, butter, and yoghurt.

3. Production of Hides and Skin:

- Their hides and skins are used for making leather products such as shoes, bags, belts, and drums.

4. Source of Income:

- Farmers sell live animals, milk, or hides to earn money, improving rural livelihoods.

5. Manure Production:

- The droppings of sheep and goats are used as organic manure to enrich farm soil.

6. Cultural and Religious Importance:

- They are used in traditional ceremonies, marriages, and religious festivals such as Eid-el-Kabir.

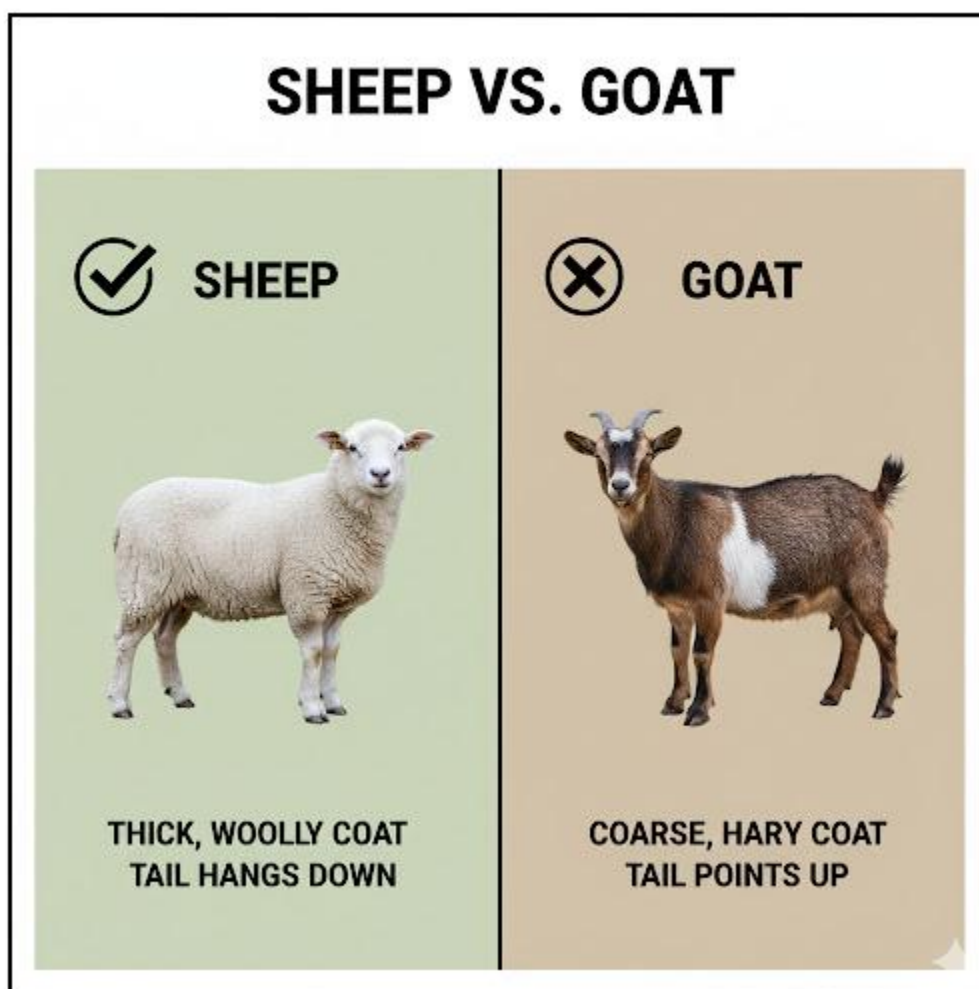
7. Employment Opportunities:

- Sheep and goat farming provides jobs for herdsman, traders, butchers, and transporters.

Differences Between Sheep and Goats

Features	Sheep	Goat
1. Lips and Browsing Habit	Prefer grazing on grass	Prefer browsing on shrubs and leaves

2. Tail Position	Tail hangs downward	Tail points upward
3. Hair/Wool	Usually covered with wool	Covered with short, coarse hair
4. Horn Shape	Usually curved or spiral horns	Usually straight or slightly curved horns
5. Behaviour	More timid and gentle	More curious and independent
6. Sound	Baa	Bleat
7. Breeding Season	Seasonal breeder	Can breed all year round



Economic Importance of Small Ruminants (Sheep and Goats)

- They serve as a **form of savings and investment** for rural farmers.
- Can be sold quickly in times of emergency or financial need.
- Require **less capital** compared to cattle.
- Contribute to **food security** and **rural development**.

EVALUATION

1. Define sheep and goat production.
2. List five importance of sheep and goat production.
3. Mention three products obtained from sheep and goats.
4. State four differences between sheep and goats.
5. Explain two economic benefits of rearing sheep and goats.

ASSIGNMENT

1. What are small ruminants? Give two examples.
2. State five uses of goats in Nigeria.
3. Mention any four materials produced from sheep and goat skin.
4. Explain why goats are more popular in northern Nigeria.
5. Draw a simple table showing three differences between sheep and goats.

WEEK 2: BREEDS OF SHEEP AND GOATS

Meaning of Breed

A **breed** is a group of animals of the same species that have similar characteristics or features such as size, color, shape, or production ability, and can pass these features to their offspring.

In livestock production, selecting the right breed is important because it determines productivity, adaptability to the environment, and disease resistance.

BREEDS OF SHEEP

Sheep breeds are classified into **local** and **exotic** breeds.

COMMON NIGERIAN BREEDS



WEST AFRICIAN
DWARF GOAT



SOKOTO RED GOAT



YANKASA SHEEP

1. Local Breeds of Sheep in Nigeria

These are sheep that have adapted to the local environment and climate. They are hardy and resistant to diseases.

Examples include:

- **West African Dwarf (WAD) Sheep:**

Found mostly in the southern part of Nigeria. They are small in size, black or brown in color, and resistant to humid conditions and diseases.

- **Yankasa Sheep:**

Commonly found in northern Nigeria. They are medium-sized, white with black patches around the eyes, ears, and legs. They are good meat producers.

- **Balami Sheep:**

Found mostly in the arid regions of northeastern Nigeria. They are large, white-coated, and well-suited for dry areas.

- **Uda (Bororo) Sheep:**

Characterized by their brown and white coat patterns. They are tall and have long legs, adapted for long-distance walking.

2. Exotic Breeds of Sheep

These are breeds imported from other countries for improved productivity.

Examples include:

- **Merino Sheep:**

Famous for fine, soft wool production.

- **Suffolk Sheep:**

Known for rapid growth and excellent mutton quality.

- **Dorset Sheep:**

Can breed all year round and produce good-quality meat and milk.

BREEDS OF GOATS

Like sheep, goat breeds are also divided into **local** and **exotic** types.

1. Local Breeds of Goats in Nigeria

Examples include:

- **West African Dwarf (WAD) Goat:**

Found in southern Nigeria; small-sized and disease-resistant. Commonly black, brown, or white.

- **Sokoto Red Goat:**

Found in the northern region; medium-sized with a red or dark-brown coat. Known for its fine skin used in leather production (*Morocco leather*).

- **Sahel (Desert) Goat:**

Tall, long-legged, and adapted to dry regions of the north. Produces more milk but less meat.

2. Exotic Breeds of Goats

Examples include:

- **Angora Goat:**

Famous for its long, silky hair called *mohair*, used in making fine clothes.

- **Boer Goat:**

Originally from South Africa; large, fast-growing, and good for meat production.

- **Saanen Goat:**

White in color and known worldwide for high milk production.

CHARACTERISTICS OF GOOD BREEDS

A good breed of sheep or goat should have the following features:

- High growth rate and productivity
- Resistance to local diseases
- Ability to adapt to the local environment
- Good mothering ability

- Strong and healthy body conformation
- High fertility and breeding capacity

USES OF DIFFERENT BREEDS

Breed	Animal Type	Main Use
West African Dwarf	Sheep/Goat	Meat, disease resistance
Yankasa	Sheep	Meat
Sokoto Red	Goat	Leather and meat
Saanen	Goat	Milk
Merino	Sheep	Wool
Boer	Goat	Meat

EVALUATION

1. Define the term "breed."
2. List three local breeds of sheep found in Nigeria.
3. Mention two exotic breeds of goats and their uses.
4. State three characteristics of a good breed.
5. Differentiate between local and exotic breeds.

ASSIGNMENT

1. Identify four local breeds of goats and their characteristics.
2. Mention two exotic breeds of sheep and describe one feature of each.

3. Draw and label two breeds of goats found in Nigeria.



WEEK 3: HOUSING AND EQUIPMENT FOR SHEEP AND GOATS

MEANING OF LIVESTOCK HOUSING

Livestock housing refers to the shelter or building provided for animals to protect them from harsh weather conditions, predators, and theft, and to ensure proper feeding, resting, and breeding.

For sheep and goats, good housing is very important because it keeps them healthy, enhances productivity, and reduces disease spread.



IMPORTANCE OF GOOD HOUSING FOR SHEEP AND GOATS

1. **Protection from weather:** Keeps animals safe from rain, cold, sun, and wind.

2. **Protection from predators:** Prevents attacks by wild animals and thieves.
3. **Disease prevention:** Ensures cleanliness and reduces infection risks.
4. **Easy management:** Makes feeding, milking, breeding, and cleaning more convenient.
5. **Improved productivity:** Comfortable animals eat well, grow faster, and produce more meat or milk.
6. **Control of parasites:** Proper housing helps in the control of ticks, lice, and flies.

TYPES OF HOUSING FOR SHEEP AND GOATS

There are different housing systems depending on the climate, location, and number of animals kept. The main types include:

1. Open Housing System

- Animals are kept in a fenced area with partial shade or simple roofing.
- Suitable for regions with dry climates.
- Animals graze freely within the compound.

2. Semi-Open Housing System

- Combines open space and a covered area.
- Provides shade and protection from rain.
- Animals are kept inside at night and allowed to graze during the day.

3. Closed or Intensive Housing System

- Animals are kept inside permanently in pens or stalls.
- Commonly used in commercial farms.
- Easier for monitoring feeding, breeding, and disease control.

REQUIREMENTS OF A GOOD SHEEP AND GOAT HOUSE

A good sheep or goat house must have the following features:

1. **Dry and Well-Drained Floor:** The floor should not retain water to prevent hoof rot and diseases.
2. **Good Ventilation:** Fresh air must circulate to remove odors and moisture.
3. **Roofing:** The roof should protect against sun and rain; materials like corrugated iron sheets, thatch, or zinc can be used.
4. **Adequate Space:** Each animal should have enough room to move and lie down.
5. **Feeding and Watering Points:** Feeders and drinkers should be properly placed.
6. **Security:** The pen must be fenced to keep predators away.
7. **Ease of Cleaning:** The floor and structure should be easy to clean and disinfect.

MATERIALS USED FOR CONSTRUCTING SHEEP AND GOAT HOUSES

- Wood or bamboo
- Cement blocks or bricks
- Corrugated iron sheets or thatch (for roofing)
- Wire mesh (for ventilation and fencing)
- Nails, ropes, and metal sheets

EQUIPMENT USED IN SHEEP AND GOAT HOUSING

Proper equipment makes management easier and improves animal welfare.

1. Feeders

- Used for holding and serving feed.

- Prevents wastage and contamination.
- Can be made from wood, metal, or concrete.

2. Water Troughs/Drinkers

- Used to provide clean drinking water.
- Should be kept clean and regularly refilled.

3. Salt Licks

- Blocks of mineralized salt placed in pens for animals to lick and obtain essential minerals.

4. Hay Racks

- Used to keep hay (dry grass) off the ground and prevent dirt contamination.

5. Milking Stand

- Used to hold the animal steady during milking.

6. Grooming Tools

- Brushes and combs for cleaning the body and removing dirt.

7. Foot Bath

- A shallow trough filled with disinfectant solution where animals step through to prevent hoof diseases.

HYGIENE AND MAINTENANCE PRACTICES

To maintain a healthy environment for sheep and goats:

- Clean pens daily to remove dung and leftover feed.
- Wash and disinfect equipment regularly.
- Change bedding materials often to keep them dry.
- Provide fresh water at all times.

- Ensure adequate drainage to prevent water stagnation.

EVALUATION

1. Define livestock housing.
2. List and explain three types of sheep and goat houses.
3. Mention five requirements of a good livestock house.
4. Identify five materials used in constructing goat houses.
5. State three reasons why cleaning livestock houses regularly is important.

ASSIGNMENT

1. Draw and label a simple goat pen showing feeding and watering points.
2. Mention four items of equipment used in sheep and goat housing and their functions.
3. Explain three advantages of proper housing to livestock farmers.
4. Describe the differences between open, semi-open, and closed housing systems.
5. Write short notes on the following:
 - a. Feeders
 - b. Drinkers
 - c. Salt licks

WEEK 4: FEEDS FOR SHEEP AND GOATS

MEANING OF FEEDS

Feeds are the different types of food materials given to animals to supply them with the nutrients necessary for growth, reproduction, maintenance, and production of meat, milk, or wool. Sheep and goats are ruminant animals — they have a four-compartment stomach (rumen, reticulum, omasum, and abomasum) — which enables them to digest roughages effectively.

Proper feeding is essential for maintaining healthy animals and achieving good production results.

RUMINANT FEEDING HABITS

GRAZING ✓



Eating short grass from the ground

BROWSING ✗



Eating leaves & shoots from shrubs

IMPORTANCE OF FEEDING SHEEP AND GOATS

1. **Promotes Growth:** Adequate feeding ensures proper body development and weight gain.

2. **Supports Reproduction:** Well-fed animals come into heat regularly and produce healthy young.
3. **Enhances Milk and Meat Production:** Feeding improves yield and quality.
4. **Maintains Health:** Balanced feed strengthens the immune system and prevents malnutrition.
5. **Improves Wool or Skin Quality:** Good nutrition enhances fiber or skin texture.

TYPES OF FEEDS FOR SHEEP AND GOATS

Feeds are grouped into two major classes:

1. Roughages

These are bulky, fibrous plant materials that ruminants eat in large quantities. They contain a lot of fiber and are the main source of feed for sheep and goats.

Examples: grasses, legumes, hay, crop residues, and silage.

Types of Roughages:

- **Green Roughages:** Fresh grass, legumes (like stylo, centro, and lucerne), and browse plants.
- **Dry Roughages:** Hay (dried grass) and straw (dried stalks of crops).
- **Silage:** Fermented and stored green fodder for use during dry season.

Importance of Roughages:

- Provide fiber for proper rumen function and digestion.
- Supply essential vitamins and minerals.
- Prevent digestive disorders.
- Are cheap and readily available.

2. Concentrates

Concentrates are energy- and protein-rich feeds given to animals in small quantities to supplement roughages. They are low in fiber but high in nutrients.

Examples: maize, wheat bran, groundnut cake, palm kernel cake, cottonseed cake, and fish meal.

Types of Concentrates:

- **Energy Concentrates:** e.g., maize, sorghum, cassava, yam peels.
- **Protein Concentrates:** e.g., groundnut cake, soybean meal, fish meal, and blood meal.

Importance of Concentrates:

- Supply energy for daily activities.
- Promote growth and reproduction.
- Help in milk and meat production.
- Improve body condition and weight.

GRAZING, BROWSING, AND SUPPLEMENTARY FEEDING

Grazing

- This is when animals feed directly on pasture grasses in open fields or paddocks.
- It is a natural and inexpensive feeding method.
- Should be done on **good quality pastures** and under **controlled stocking** to avoid overgrazing.

Browsing

- This involves goats and sometimes sheep feeding on leaves, shoots, and twigs of shrubs and trees.
- Goats are natural browsers and prefer leaves from plants like leucaena, gliricidia, and acacia.

- Browsing provides vitamins and minerals.

Supplementary Feeding

- This is the additional feeding of concentrates, salt licks, or crop residues to animals when grazing is insufficient (especially during the dry season).
- Supplements are essential for pregnant, lactating, or young animals that need extra nutrients.

FEEDING PRACTICES FOR SHEEP AND GOATS

1. Feed animals **twice daily** — morning and evening.
2. Provide **clean and fresh water** at all times.
3. Offer **salt licks** to supply minerals.
4. Avoid feeding **wet, mouldy, or dirty feed**.
5. Ensure animals are not **overfed with concentrates** to prevent digestive problems.
6. Feed according to **age, body size, and production stage**.
7. Rotate grazing areas to allow pastures to regrow.

IMPORTANCE OF CLEAN WATER

Water is essential for digestion, temperature regulation, and general body functions.

- Sheep and goats should have **constant access** to clean, cool water.
- Dirty water may spread diseases and parasites.
- During dry seasons, ensure water is supplied at least **twice a day**.

EVALUATION

1. Define feed as it relates to sheep and goats.
2. Mention and explain two major types of feeds for sheep and goats.
3. State four examples of roughages and four examples of concentrates.
4. Explain the difference between grazing and browsing.
5. Give five good feeding practices for sheep and goats.

ASSIGNMENT

1. Explain three reasons why feeding is important in sheep and goat farming.
2. List five examples each of energy and protein concentrates.
3. Describe three methods of feeding ruminant animals.
4. Write short notes on:
 - a. Silage
 - b. Hay
 - c. Salt lick
5. Prepare a one-day feeding schedule for a goat that includes roughages, concentrates, and water.

WEEK 6: HEALTH MANAGEMENT IN SHEEP AND GOATS

MEANING OF HEALTH MANAGEMENT

Health management refers to all the practices and activities carried out to maintain the good health and wellbeing of sheep and goats. It includes preventive measures, treatment, hygiene, and general care aimed at reducing diseases, improving productivity, and ensuring long life.

Healthy sheep and goats grow well, reproduce easily, and yield more meat, milk, or wool. Poor health leads to low productivity, losses, and death.

IMPORTANCE OF HEALTH MANAGEMENT

1. **Prevents Diseases:** Helps to reduce the occurrence and spread of diseases and parasites.
2. **Improves Productivity:** Healthy animals grow faster and reproduce better.
3. **Reduces Mortality:** Fewer animals die when good health care is practiced.
4. **Increases Profit:** Healthy livestock ensures better returns to the farmer.
5. **Ensures Food Safety:** Healthy animals produce safe meat, milk, and skin for human use.
6. **Promotes Animal Welfare:** Ensures animals live comfortably and free from pain or distress.

COMMON DISEASES AND PARASITES OF SHEEP AND GOATS

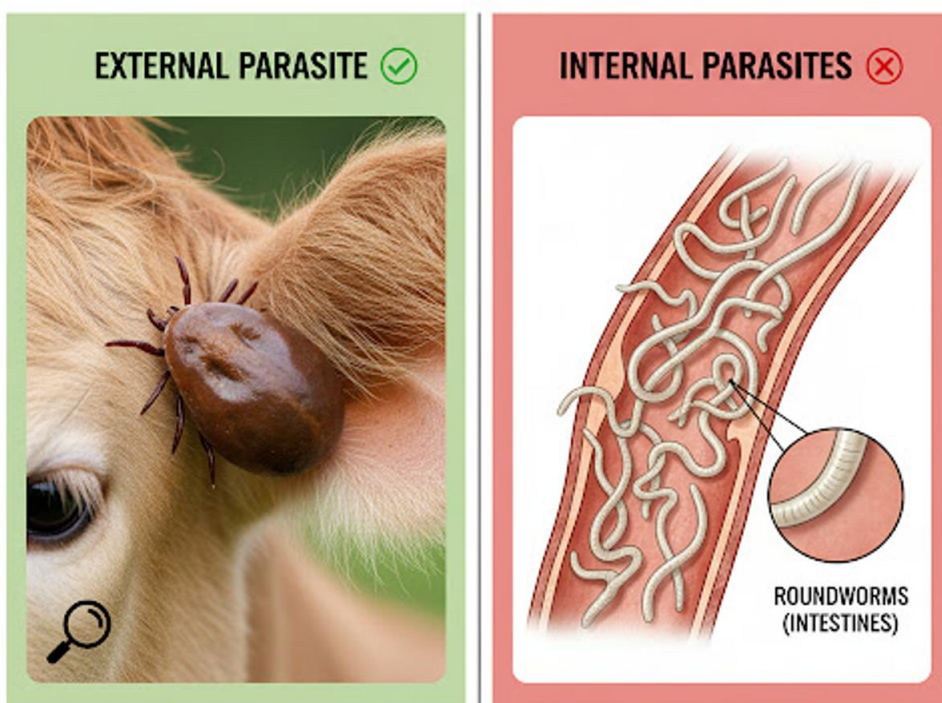
Sheep and goats can be affected by several diseases, which may be **infectious** (spread from animal to animal) or **non-infectious** (caused by poor feeding or management).

1. COMMON DISEASES

Disease	Causes	Symptoms	Prevention and Control
Peste des Petits Ruminants (PPR)	Virus	Fever, coughing, nasal discharge, diarrhea, death	Vaccinate animals regularly; isolate sick animals

Foot Rot	Bacteria	Lameness, foul smell, swollen hooves	Keep floor dry, use foot bath with disinfectant
Mastitis	Bacteria	Swollen udder, abnormal milk	Keep milking area clean, treat with antibiotics
Pneumonia	Cold and infection	Coughing, breathing difficulty	Provide warm shelter, avoid cold drafts
Bloat	Excess gas in rumen	Swollen abdomen, discomfort	Avoid wet grasses; give anti-bloat medicine

LIVESTOCK PARASITES



2. COMMON PARASITES

Parasites are organisms that live on or inside the animal's body and cause harm. They may be **internal** or **external**.

(a) Internal Parasites (Endoparasites)

- Found inside the animal's body, mainly in the stomach or intestines.
- Examples: roundworms, tapeworms, liver flukes.
- **Symptoms:** loss of weight, rough coat, diarrhea, poor growth.
- **Control:** deworm regularly (every 3 months), keep pens clean, avoid overgrazing.

(b) External Parasites (Ectoparasites)

- Found on the skin or wool of the animal.
- Examples: ticks, lice, mites, fleas.
- **Symptoms:** scratching, hair loss, wounds, restlessness.
- **Control:** spray or dip animals with recommended insecticides, clean housing regularly.

DISEASE PREVENTION AND CONTROL METHODS

1. **Vaccination:** Give animals vaccines against deadly diseases like PPR and anthrax.
2. **Regular Deworming:** Use dewormers to remove internal parasites every few months.
3. **Isolation (Quarantine):** Keep new or sick animals away from healthy ones.
4. **Good Housing:** Keep pens dry, clean, and well-ventilated.
5. **Proper Feeding:** Feed animals balanced diets to strengthen their immune systems.
6. **Clean Water Supply:** Provide fresh and uncontaminated water.
7. **Personal Hygiene:** Farmers should wash hands and disinfect tools after handling animals.
8. **Record Keeping:** Keep records of vaccinations, treatments, and diseases for future reference.

SIGNS OF HEALTHY SHEEP AND GOATS

- Bright and alert eyes

- Smooth and shiny coat
- Good appetite
- Normal temperature and breathing
- Active movement
- Clean nose and mouth
- Regular droppings

SIGNS OF UNHEALTHY SHEEP AND GOATS

- Dull or sunken eyes
- Rough coat or loss of hair
- Loss of appetite
- Weakness or isolation from the flock
- Coughing, sneezing, or diarrhea
- Swelling or wounds on the body

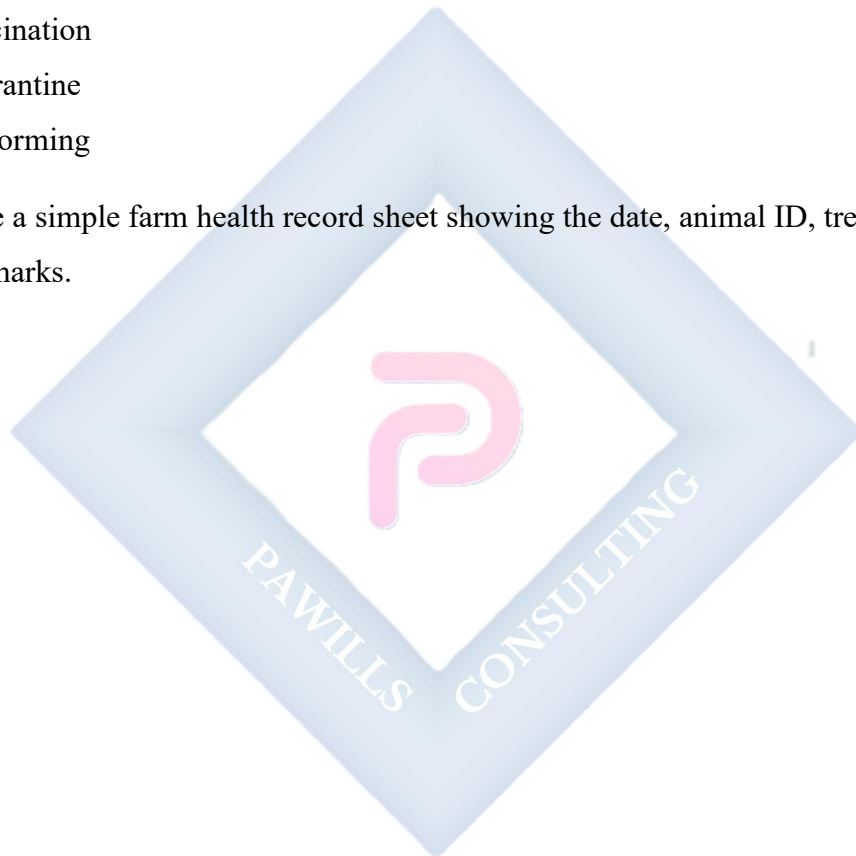
When such signs are noticed, **the farmer must quickly isolate the animal and call a veterinarian.**

EVALUATION

1. Define health management.
2. State four importance of good health management in livestock.
3. Mention three common diseases of sheep and goats and their symptoms.
4. Differentiate between internal and external parasites, giving two examples each.
5. List five preventive measures for maintaining healthy sheep and goats.

ASSIGNMENT

1. Explain three ways to prevent disease outbreaks in a goat farm.
2. Draw and label one internal and one external parasite of goats.
3. List five signs of a healthy goat.
4. Write short notes on:
 - a. Vaccination
 - b. Quarantine
 - c. Deworming
5. Prepare a simple farm health record sheet showing the date, animal ID, treatment given, and remarks.



WEEK 8: SICK SHEEP OR GOAT IN A HERD

MEANING OF A SICK ANIMAL

A sick animal is one that is not in good health due to the presence of disease, infection, injury, or poor management. In a herd, when one animal becomes sick, it poses a risk to the rest if not identified and treated promptly. Early detection and care of sick animals help prevent the spread of diseases and loss of livestock.



CAUSES OF SICKNESS IN SHEEP AND GOATS

1. **Poor feeding:** Lack of balanced nutrition weakens the immune system.
2. **Dirty housing conditions:** Accumulation of waste encourages germs and parasites.
3. **Contaminated water:** Leads to infections and digestive problems.
4. **Poor sanitation and hygiene:** Promotes spread of bacteria, viruses, and parasites.

5. **Lack of vaccination and deworming:** Makes animals susceptible to diseases.
6. **Extreme weather conditions:** Cold, rain, or heat stress weakens the animals.
7. **Injuries and wounds:** Allow germs to enter the body.

SIGNS OF A SICK SHEEP OR GOAT

Recognizing early symptoms of illness helps the farmer act quickly.

Common signs include:

1. Dull or sunken eyes
2. Rough, dull, or loss of hair coat
3. Loss of appetite and weight loss
4. Isolation from other animals
5. Coughing, sneezing, or nasal discharge
6. Diarrhea or constipation
7. Swollen body parts (udder, joints, or stomach)
8. Restlessness or weakness
9. Limping or difficulty in movement
10. Decrease in milk production in lactating animals

IDENTIFICATION OF SICK ANIMALS IN A HERD

To identify sick animals effectively:

1. **Observe animals daily:** Check for signs of unusual behavior or appearance.
2. **Monitor feeding habits:** Sick animals usually refuse to eat or drink.
3. **Check body condition:** Feel for abnormal body temperature or swelling.

4. **Examine droppings:** Watery or discolored feces indicate illness.
5. **Keep health records:** Compare current performance to past behavior.

Regular observation during feeding or cleaning helps farmers detect sickness early before it spreads.

ISOLATION OF SICK ANIMALS

Isolation means separating sick animals from healthy ones to prevent the spread of contagious diseases.

Steps to follow:

1. Move the sick animal immediately to a separate pen or quarantine area.
2. Provide clean feed, water, and bedding.
3. Limit contact between sick and healthy animals.
4. Disinfect the area after handling the sick animal.
5. Call a veterinarian for diagnosis and treatment.

Isolation pens should be located **downwind** (away from the main herd) and should have **dedicated equipment** not used by healthy animals.

TREATMENT AND CARE OF SICK ANIMALS

Once a sick animal is identified and isolated, the following steps should be taken:

1. **Consult a Veterinarian:** Get professional help for proper diagnosis and medication.
2. **Administer Treatment:** Follow the vet's advice on drugs, dosage, and duration.
3. **Provide Balanced Feeding:** Give nutritious feed and clean water to aid recovery.
4. **Maintain Clean Environment:** Keep the isolation area clean and dry.
5. **Observe Progress:** Record the animal's temperature, appetite, and recovery daily.

6. **Disinfect Tools and Hands:** Wash and disinfect after handling sick animals.
7. **Dispose of Dead Animals Properly:** If death occurs, bury or burn the carcass to avoid contamination.

RECORD KEEPING FOR SICK ANIMALS

Maintaining proper records helps in monitoring the health history of each animal and prevents repeated outbreaks.

Essential health records include:

- Date of illness
- Animal identification number (ear tag or name)
- Symptoms observed
- Type of treatment or drug administered
- Dosage and duration
- Response to treatment
- Remarks or outcome (recovery or death)

These records guide future management decisions and help the vet in follow-up care.

PREVENTION OF SICKNESS IN THE HERD

1. Provide balanced feed and clean water daily.
2. Keep pens dry and disinfect regularly.
3. Carry out regular deworming and vaccination.
4. Isolate new animals before adding them to the herd.
5. Remove sharp objects to prevent injuries.

6. Practice good hygiene among workers and handlers.
7. Avoid overcrowding in pens.

EVALUATION

1. Define a sick animal.
2. List five signs of sickness in sheep and goats.
3. Mention four causes of sickness in livestock.
4. Explain three reasons for isolating sick animals.
5. State five preventive measures to reduce sickness in a herd.

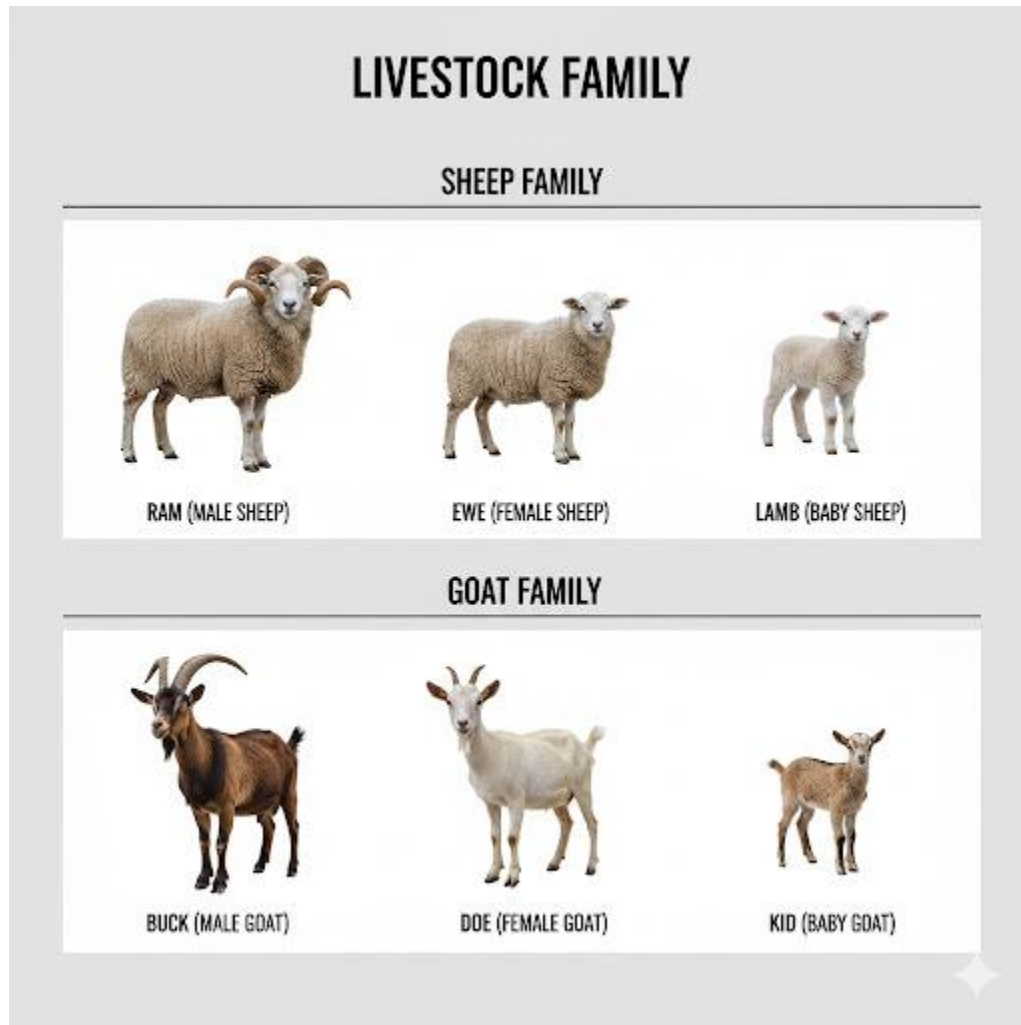
ASSIGNMENT

1. Write short notes on:
 - a. Isolation
 - b. Treatment
 - c. Disinfection
2. Outline six steps to take when a goat suddenly becomes sick.
3. Mention four reasons why record keeping is important in livestock health management.
4. Describe three signs that show an animal is recovering from sickness.
5. Design a simple “Animal Health Record Form” with spaces for animal ID, symptoms, treatment, and outcome.

WEEK 9: BREEDING IN SHEEP AND GOATS

MEANING OF BREEDING

Breeding is the process of mating male and female animals to produce young ones. In sheep and goats, breeding ensures the multiplication of stock, improvement of desired traits, and increased productivity (meat, milk, or wool). Breeding can occur naturally through mating or artificially through the use of semen in artificial insemination.



TYPES OF BREEDING

1. Natural Breeding:

This is when the male and female animals mate directly. The male mounts the female to deposit sperm during heat (estrus).

- **Advantages:** Simple, inexpensive, and requires little skill.
- **Disadvantages:** Disease transmission risk; difficult to control breeding timing.

2. **Artificial Breeding (Artificial Insemination):**

This involves collecting semen from a healthy male and inserting it into the female's reproductive tract using special equipment.

- **Advantages:** Prevents spread of diseases, allows use of superior male semen for many females, and improves breeding records.
- **Disadvantages:** Requires skill, equipment, and may be costly.

BREEDING TERMS AND DEFINITIONS

- **Ram:** A mature male sheep.
- **Ewe:** A mature female sheep.
- **Buck (or Billy):** A mature male goat.
- **Doe (or Nanny):** A mature female goat.
- **Kid:** A young goat.
- **Lamb:** A young sheep.
- **Mating (Service):** The act of the male mounting the female for reproduction.
- **Gestation:** The period from conception to birth (pregnancy period).
- **Parturition:** The act of giving birth.
- **Weaning:** The process of stopping young animals from suckling.

GESTATION PERIODS

- **Sheep: About 150 days (5 months)**

- Goats: **About 150 days (5 months)**

During this period, the pregnant female should be handled carefully, fed nutritious feed, and kept in a clean, quiet pen.

SIGNS OF HEAT IN SHEEP AND GOATS

A female ready for breeding shows the following signs:

1. Restlessness and bleating frequently
2. Swollen and reddish vulva
3. Mucus discharge from the vulva
4. Frequent wagging of the tail
5. Standing still when the male mounts
6. Loss of appetite or nervous behavior

Heat lasts for **about 1–2 days** and occurs every **18–21 days** if the animal is not mated.

BREEDING PRACTICES

1. Selection of Breeding Stock:

- Choose healthy, strong, and disease-free animals.
- Select animals with desirable traits such as fast growth, good milk yield, and fertility.
- Avoid inbreeding (mating close relatives) to prevent weak offspring.

2. Mating Management:

- Observe females for heat signs and allow mating during this period.

- Ensure that only one healthy male serves selected females to maintain breed quality.
- Record the date of mating and expected delivery date.

3. Care of Pregnant Animals:

- Provide adequate feed and clean water.
- Avoid overworking or stressing the animal.
- Keep the animal in a clean, well-ventilated pen.
- Vaccinate and deworm under veterinary supervision.

4. Care at Parturition (During Birth):

- Ensure a quiet, clean area for delivery.
- Observe for signs of labor (restlessness, isolation, straining).
- Assist only when necessary or call a veterinarian.
- Clean the newborn and ensure it suckles colostrum within the first few hours.

5. Post-Birth Care:

- Disinfect the navel of the newborn.
- Provide warmth and good bedding.
- Feed the mother with nutritious feed to enhance milk production.

RECORD KEEPING IN BREEDING

It is important to maintain accurate breeding records such as:

- Animal identification (name or tag number)
- Date of mating and expected date of parturition
- Date of birth and number of offspring born

- Sex and weight of young ones
- Remarks on health and survival rate

These records help farmers monitor reproductive performance and improve breeding programs.

IMPORTANCE OF BREEDING IN SHEEP AND GOATS

1. Increases the number of animals in the herd
2. Improves quality of meat, milk, and wool
3. Enhances income generation for farmers
4. Preserves desirable traits within a breed
5. Allows for better planning of production cycles

EVALUATION

1. Define breeding.
2. State two types of breeding and explain each.
3. List three signs of heat in a female goat or sheep.
4. What is the gestation period of a goat?
5. Mention five good qualities of a breeding stock.
6. Differentiate between natural and artificial breeding.

ASSIGNMENT

1. Explain the term “parturition” and state three signs that a ewe is about to give birth.
2. Mention and describe four practices that ensure successful breeding in goats.
3. Draw and label a simple table showing:

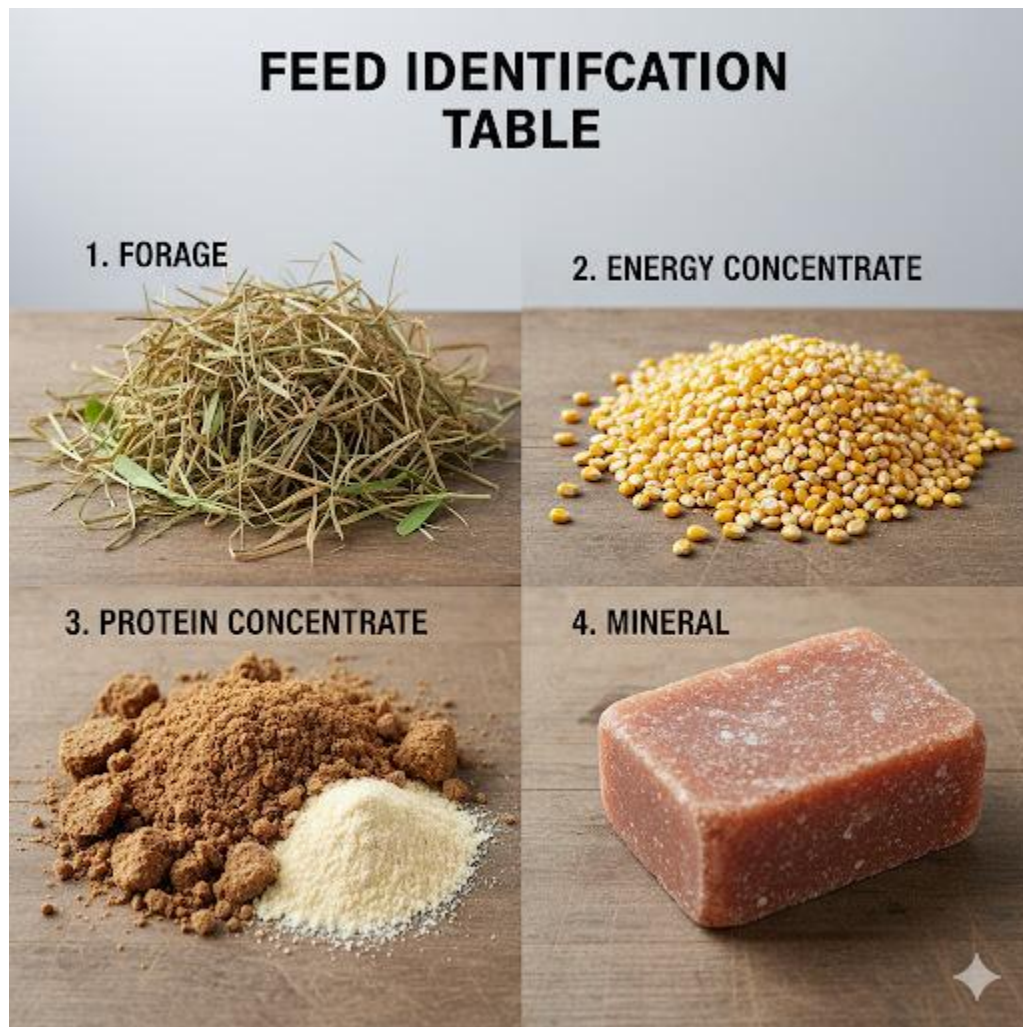
- Animal name
- Mating date
- Expected delivery date
- Number of offspring produced



WEEK 10: PRACTICAL – IDENTIFYING FEEDS AND BREEDS

INTRODUCTION

Practical knowledge is essential in livestock farming because it helps students and farmers apply theoretical lessons to real-life situations. In this lesson, students will learn how to identify **different breeds of sheep and goats** and **various types of feeds** commonly used for their nutrition. This practical session enhances observation, record-keeping, and classification skills in animal husbandry.



PART A: IDENTIFYING BREEDS OF SHEEP AND GOATS

1. COMMON BREEDS OF SHEEP

1. West African Dwarf (WAD) Sheep

- **Description:** Small-sized, short legs, and coat colour ranges from white to brown or black.
- **Adaptation:** Very hardy and well-suited to humid regions of West Africa.
- **Uses:** Meat and skin production.

2. Yankasa Sheep

- **Description:** Medium to large size, white body with black patches around the eyes and muzzle, spiral horns in males.
- **Adaptation:** Common in the savanna region of Nigeria.
- **Uses:** Meat, skin, and for cultural purposes.

3. Balami Sheep

- **Description:** Large-sized, entirely white coat, droopy ears, and long legs.
- **Adaptation:** Best suited for dry and semi-arid regions.
- **Uses:** Meat and wool production.

2. COMMON BREEDS OF GOATS

1. West African Dwarf (WAD) Goat

- **Description:** Small size, short legs, variable coat colour, hardy and disease-resistant.
- **Adaptation:** Thrives in humid forest zones.
- **Uses:** Meat production and domestic purposes.

2. Sokoto Red (Maradi) Goat

- **Description:** Medium-sized with a shiny reddish-brown coat.
- **Adaptation:** Common in northern Nigeria.

- **Uses:** Skin for leather (Morocco leather), and meat.

3. Sahel (Desert) Goat

- **Description:** Tall and lean with long legs, mostly white or light brown.
- **Adaptation:** Suited for arid and semi-arid areas.
- **Uses:** Milk and meat production.

4. Nubian Goat

- **Description:** Large-sized with long droopy ears and convex face; coat may be brown, black, or mixed.
- **Adaptation:** Performs well in both tropical and temperate climates.
- **Uses:** Dual-purpose — milk and meat.

PART B: IDENTIFYING TYPES OF FEEDS FOR SHEEP AND GOATS

Sheep and goats require different categories of feeds to remain healthy and productive. Feeds can be identified and classified as follows:

1. FORAGES (Roughages)

These are plant materials eaten by animals in large quantities, usually high in fibre but low in nutrients.

- **Examples:**

- Fresh grasses (Guinea grass, Elephant grass)
- Leguminous plants (Centrosema, Stylo, Calopogonium)
- Dry grasses (Hay)
- Dried leaves and crop residues

2. CONCENTRATES

Feeds that contain high nutrients (protein, carbohydrate, fat) and low fibre.

- **Types and Examples:**

- **Energy concentrates:** Maize, cassava peels, sorghum, yam peels.
- **Protein concentrates:** Groundnut cake, soybean meal, cottonseed cake, fish meal, blood meal.

3. MINERAL SUPPLEMENTS

Provide essential minerals like calcium, phosphorus, and salt.

- **Examples:** Bone meal, salt lick, oyster shell powder.

4. WATER

Clean and fresh water is vital for digestion, milk production, and general health. Goats and sheep should have constant access to clean water.

PRACTICAL ACTIVITIES

1. Breed Identification Activity:

- Observe live sheep and goats or pictures/models.
- Note distinguishing features (size, coat colour, ear shape, and horn type).
- Classify each animal into its correct breed group.

2. Feed Identification Activity:

- Display samples of different feed ingredients.
- Students will identify, touch, and group them as forage, concentrate, or mineral supplement.

3. Recording Observations:

- Use a table to record breed name, characteristics, and feed category.

Example of a Record Table:

Animal/Breed	Distinctive Features	Feed Type	Feed Example
WAD Goat	Small size, short legs, black coat	Roughage	Elephant grass
Sokoto Red Goat	Shiny red coat, medium size	Concentrate	Groundnut cake
Yankasa Sheep	White body, black face patches	Mineral	Salt lick

IMPORTANCE OF IDENTIFYING FEEDS AND BREEDS

1. Helps in proper feeding and nutrition planning.
2. Assists in selecting suitable breeds for specific environments.
3. Promotes better management and productivity.
4. Prevents feeding errors that may cause diseases.
5. Improves farmers' practical knowledge of animal characteristics.

EVALUATION

1. Mention four breeds of goats and describe one distinctive feature of each.
2. List and explain three categories of sheep and goat feeds.
3. State three reasons why breed identification is important.
4. Differentiate between forage and concentrate with examples.
5. What is the major importance of mineral supplements in animal feeding?

ASSIGNMENT

1. Prepare a table showing at least **five feed ingredients** grouped under roughage, concentrate, and mineral supplement.

2. Draw and label **two common breeds** of sheep and goats found in Nigeria.
3. Explain how proper breed identification can improve livestock productivity in your locality.



WEEK 11: REVISION – REVIEW OF TERM'S WORK AND KEY LIVESTOCK PRACTICES

INTRODUCTION

This week's lesson is dedicated to revising all the topics studied during the third term. The revision will focus on reinforcing students' understanding of **sheep and goat production**, their **housing, feeding, breeding, and health management**. This session ensures students recall important facts, apply learned skills, and prepare adequately for the end-of-term examination.

REVIEW OF KEY TOPICS

1. CONCEPT OF SHEEP AND GOAT PRODUCTION

- **Meaning:**

Sheep and goat production refers to the rearing and management of these animals for food, income, clothing materials, and other by-products.

- **Importance:**

- Source of **meat, milk, leather**, and **manure**.
- Provides **income** and **employment** opportunities.
- Contributes to **cultural and traditional** practices.
- Helps in **weed control** through grazing.

- **Differences between Sheep and Goats:**

- Sheep have **woolly coats** while goats have **hairy coats**.
- Sheep usually **graze**, while goats **browse**.
- Male sheep have a **strong odour**, while goats' odour is milder.
- Sheep's tail points **downward**, goats' tail points **upward**.

2. BREEDS OF SHEEP AND GOATS

- **Common Breeds of Sheep:**
 - **West African Dwarf Sheep:** Small size, hardy, found in southern Nigeria.
 - **Yankasa Sheep:** Medium to large, white with black patches, found in northern Nigeria.
 - **Balami Sheep:** Large white coat, suited to dry regions.
- **Common Breeds of Goats:**
 - **West African Dwarf Goat:** Small, hardy, good for humid areas.
 - **Sokoto Red Goat:** Reddish coat, produces fine leather.
 - **Sahel Goat:** Tall and lean, suited to arid zones.
 - **Nubian Goat:** Large dual-purpose breed for milk and meat.

3. HOUSING AND EQUIPMENT FOR SHEEP AND GOATS

- **Types of Houses:**
 - **Shed type:** Open structure with roofing, suitable for ventilation.
 - **Pen type:** Fenced area with a small shelter.
- **Requirements of a Good House:**
 - Clean and dry floor.
 - Proper drainage system.
 - Adequate ventilation.
 - Sufficient space for movement.
- **Equipment:**

Feeders, drinkers, troughs, hay racks, and mineral lick stones.

4. FEEDS FOR SHEEP AND GOATS

- **Forages:** Grasses, legumes, hay, and crop residues.
- **Concentrates:** Groundnut cake, maize, cassava peels, and fish meal.
- **Minerals:** Bone meal, salt lick, and oyster shell powder.
- **Water:** Must be clean, cool, and available always.

Feeding Methods:

- **Grazing:** Animals feed on natural pastures.
- **Browsing:** Animals eat leaves, shrubs, and twigs.
- **Supplementary Feeding:** Feed given to improve nutrition balance.

5. HEALTH MANAGEMENT IN SHEEP AND GOATS

- **Common Diseases and Parasites:**
 - Foot rot, pneumonia, internal worms, lice, and ticks.
 - **Symptoms:** Loss of appetite, coughing, limping, diarrhoea, dullness.
 - **Prevention:**
 - Regular cleaning and disinfection of pens.
 - Vaccination and deworming.
 - Isolating sick animals.
 - Proper feeding and watering.

6. BREEDING AND REPRODUCTION

- **Mating Systems:** Natural mating and controlled mating.
- **Gestation Period:** About **150 days (5 months)** for both sheep and goats.

- **Kindling/Lambing:** Act of giving birth in goats/sheep respectively.
- **Care of Young:** Provide warmth, milk, and a clean environment.

7. PRACTICAL ACTIVITIES REVISED

Students should be able to:

1. Identify various breeds of sheep and goats.
2. Classify feed ingredients into roughages, concentrates, and minerals.
3. Demonstrate proper feeding and watering methods.
4. Recognize signs of disease and apply preventive measures.
5. Keep simple records of feeding, breeding, and health practices.

SUMMARY OF TERM'S WORK

The third term focused on:

1. Understanding the concept, importance, and breeds of sheep and goats.
2. Learning housing, feeding, and health management systems.
3. Studying reproduction and breeding processes.
4. Engaging in practical identification of feeds and breeds.
5. Preparing for independent management of small livestock units.

EVALUATION

1. State four differences between sheep and goats.
2. Mention three common breeds of goats and describe their characteristics.
3. List five equipment used in sheep and goat farming.

4. Explain three methods of feeding sheep and goats.
5. Identify three common diseases of goats and their preventive measures.

ASSIGNMENT

1. Prepare a short note on the importance of housing in sheep and goat farming.
2. Draw and label a simple pen suitable for housing goats.
3. Outline five ways to maintain good animal health.
4. Create a table showing three breeds of sheep and three breeds of goats with their uses.
5. Write out five lessons you learned this term and how they can help you become a better livestock farmer.

